

## Financial Management -- Important Topics & Questions (Exam-Oriented)

### Unit I -- Introduction to Financial Management

✓ Important Topics • Nature, Scope & Functions of Financial Management • Objectives of Financial Management • Profit Maximization vs Wealth Maximization • Role of Finance Manager • Types of Business Finance -- Equity, Debt, Internal Financing • Importance of Finance for Engineers & Startups

#### ? Important Questions

1. Define Financial Management and explain its nature and scope.
2. Explain the functions of Financial Management.
3. Compare Profit Maximization and Wealth Maximization.
4. Explain the role and responsibilities of a Finance Manager.
5. Discuss various sources of business finance.
6. Explain Equity, Debt and Internal Financing.
7. Why is financial management important for engineers and startups?
8. Differentiate between Equity Financing and Debt Financing.

### Unit II -- Time Value of Money & Capital Budgeting

✓ Important Topics • Time Value of Money (TVM) • Present Value (PV) & Future Value (FV) • Payback Period • Net Present Value (NPV) • Internal Rate of Return (IRR) • Accounting Rate of Return (ARR) • Profitability Index (PI) • Engineering Project Evaluation

#### ? Important Questions

1. Explain the concept and importance of Time Value of Money.
2. Derive the formulas of Present Value and Future Value.
3. Calculate PV and FV. (Numerical)
4. Explain the Payback Period method with an example.
5. Explain the NPV method. (Numerical)
6. Explain the IRR method.
7. Explain the ARR method.
8. Explain the PI method.
9. Compare NPV and IRR methods.
10. Evaluate an engineering project using capital budgeting techniques. (Numerical)

### Unit III -- Financial Statements & Ratio Analysis

✓ Important Topics • Profit & Loss Account • Balance Sheet • Cash Flow Statement • Liquidity Ratios • Profitability Ratios • Solvency Ratios • Turnover Ratios • Interpretation of Annual Reports using Ratio Analysis

#### ? Important Questions

1. Explain the components of Financial Statements.
2. Differentiate between P&L Account and Balance Sheet.
3. Explain the Cash Flow Statement.
4. Explain Liquidity Ratios with formulas.
5. Explain Profitability Ratios with formulas.
6. Explain Solvency Ratios with formulas.

7. Explain Turnover Ratios with formulas.
8. Perform ratio analysis for a given financial statement. *(Numerical)*
9. How can annual reports be interpreted using ratio analysis?
10. Write short notes on: • Current Ratio • Quick Ratio • Debt-Equity Ratio • Inventory Turnover Ratio • Return on Investment (ROI)

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 *Unit IV -- Applications*

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♦ *Working Capital Management -- Important Topics* • Concept and Importance of Working Capital • Components of Working Capital • Inventory, Receivables & Payables • Operating Cycle • Estimation of Working Capital • Working Capital Requirement Numericals • Applications in Manufacturing & Service Industries

 *Important Questions*

1. Define Working Capital and explain its importance.
2. Explain the components of Working Capital.
3. Explain the Operating Cycle with a diagram.
4. Explain the estimation of Working Capital.
5. Calculate Working Capital Requirement. *(Numerical)*
6. Explain Working Capital Management in Manufacturing Industries.
7. Explain Working Capital Management in Service Industries.

♦ *Sources of Finance & Cost of Capital -- Important Topics* • Classification of Sources of Finance • Short-Term & Long-Term Finance • Equity Shares • Preference Shares • Debt Financing • Retained Earnings • Cost of Capital • Weighted Average Cost of Capital (WACC) • Role of Banks, Financial Institutions & Capital Markets

 *Important Questions*

1. Classify the sources of finance.
2. Differentiate between Short-Term and Long-Term sources of finance.
3. Explain Equity Shares and Preference Shares.
4. Explain Debt Financing and Retained Earnings.
5. Define Cost of Capital.
6. Explain the concept of WACC.
7. Calculate WACC for a given capital structure. *(Numerical)*
8. Explain the role of banks and financial institutions in financing businesses.
9. Explain the role of capital markets in raising finance.
10. Write short notes on: • Retained Earnings • Preference Shares • Cost of Capital • WACC

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 *Most Important for Exam (High Probability)*

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 *Theory* ✓ Profit Maximization vs Wealth Maximization ✓ Functions of Financial Management ✓ Role of Finance Manager ✓ Financial Statements ✓ Ratio Analysis ✓ Working Capital & Operating Cycle ✓ Sources of Finance ✓ WACC

 *Numericals* ✓ Present Value (PV) & Future Value (FV) ✓ Payback Period ✓ NPV ✓ IRR ✓ ARR ✓ Profitability Index (PI) ✓ Working Capital Requirement ✓ Ratio Analysis ✓ WACC

 *Financial Accounting -- Important Topics & Questions (Exam-Oriented)*

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 *Unit I -- Accounting for Assets*

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✓ *Important Topics* • Property, Plant & Equipment (PP&E) • Acquisition, Cost Determination & Capitalization • Depreciation Methods:

- Straight Line Method (SLM)

- Reducing Balance Method (RBM)
- Units of Production Method • Impairment of Assets • Disposal and Exchange of Assets • Accounting Standards: AS 10, IAS 16
- Inventory Costing Methods:
  - FIFO
  - LIFO
  - Weighted Average Method • Lower of Cost or Net Realizable Value (LCNRV) • Periodic and Perpetual Inventory Systems • Accounting Standards: AS 2, IAS 2
- Intangible Assets:
  - Recognition & Measurement
  - Amortization
  - Goodwill, Patents, Trademarks
  - Impairment
  - Research & Development Costs • Accounting Standards: AS 26, IAS 38

### *Important Questions*

1. Explain the accounting treatment of Property, Plant and Equipment (PP&E).
2. Explain the methods of depreciation with examples.
3. Differentiate between SLM and RBM.
4. What is impairment of assets? Explain its accounting treatment.
5. Explain accounting for disposal and exchange of assets.
6. Explain FIFO, LIFO and Weighted Average methods.
7. Explain LCNRV with an example.
8. Differentiate between Periodic and Perpetual Inventory Systems.
9. Explain recognition and amortization of intangible assets.
10. Explain accounting treatment of research and development costs.
11. Write short notes on: • Goodwill • Patents • Trademarks • AS 10 and AS 26

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### *Unit II -- Accounting for Liabilities and Equity*

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### *Important Topics* • Current Liabilities:

- Accounts Payable
- Short-Term Debt
- Long-Term Liabilities:
  - Bonds Payable
  - Loans
- Provisions, Contingent Liabilities & Contingent Assets • Accounting Standards: AS 29, IAS 37
- Share Capital:
  - Types of Shares
  - Issue of Shares
  - Forfeiture and Reissue of Shares
  - Buyback of Shares
  - Bonus Shares
- Dividends:

- Interim Dividend
- Final Dividend
- Declaration & Payment

### ? Important Questions

1. Explain accounting treatment of current liabilities.
2. Explain accounting treatment of long-term liabilities.
3. Differentiate between provisions and contingent liabilities.
4. Explain contingent assets with examples.
5. Explain various types of shares.
6. Explain accounting entries for issue of shares.
7. Explain forfeiture and reissue of shares.
8. Explain share buyback and bonus shares.
9. Explain interim and final dividends.
10. Write short notes on: • Provisions • Contingent Liabilities • Share Capital • Dividends

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### Unit III -- Corporate Accounting

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✓ Important Topics • Accounting for Share Capital • Accounting for Debentures • Company Final Accounts • Cash Flow Statement (AS 3 / IAS 7) • Share Buyback • Bonus Shares and Rights Issue • Interim and Final Dividends • Branch Accounting:

- Dependent Branch
- Independent Branch
- Lease Accounting:
  - Operating Lease
  - Finance Lease

### ? Important Questions

1. Explain accounting for share capital.
2. Explain accounting treatment of debentures.
3. Prepare company final accounts. (Numerical)
4. Explain the concept of Cash Flow Statement.
5. Prepare a Cash Flow Statement. (Numerical)
6. Explain accounting treatment of share buyback.
7. Explain bonus shares and rights issue.
8. Differentiate between interim and final dividends.
9. Explain dependent and independent branch accounting.
10. Differentiate between operating lease and finance lease.
11. Write short notes on: • Debentures • Rights Issue • Cash Flow Statement • Lease Accounting

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### Unit IV -- Special Accounting Topics

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✓ Important Topics • Objectives of Financial Statement Analysis • Horizontal Analysis (Trend Analysis) • Vertical Analysis (Common Size Statements) • Ratio Analysis:

- Liquidity Ratios
- Solvency Ratios
- Profitability Ratios
- Efficiency Ratios
- Interpretation of Financial Statements • Limitations of Financial Statement Analysis:

- Historical Cost Concept
- Use of Estimates
- Different Accounting Policies
- Non-Financial Factors
- Manipulation of Statements

### *Important Questions*

1. Explain the objectives of financial statement analysis.
2. Explain Horizontal Analysis with an example.
3. Explain Vertical Analysis with an example.
4. Explain various ratios used in financial statement analysis.
5. Explain liquidity, solvency, profitability and efficiency ratios.
6. Explain interpretation of financial statements.
7. Discuss the limitations of financial statement analysis.
8. Perform ratio analysis for a given financial statement. *(Numerical)*
9. Differentiate between Horizontal and Vertical Analysis.
10. Write short notes on: • Trend Analysis • Common Size Statements • Ratio Analysis • Limitations of Financial Statement Analysis



### *Most Important for Exam (High Probability)*

 *Theory* ✓ Depreciation Methods ✓ Impairment of Assets ✓ Inventory Costing Methods (FIFO, LIFO, Weighted Average) ✓ Intangible Assets ✓ Provisions & Contingent Liabilities ✓ Share Capital & Debentures ✓ Cash Flow Statement ✓ Branch Accounting ✓ Lease Accounting ✓ Financial Statement Analysis & Ratio Analysis

 *Numericals* ✓ Depreciation (SLM & RBM) ✓ Inventory Valuation (FIFO, LIFO, Weighted Average) ✓ Company Final Accounts ✓ Cash Flow Statement ✓ Ratio Analysis

 *Digital Finance -- Important Topics & Questions (Exam-Oriented)*



### *Unit I -- Digital Finance Fundamentals & Big Data*

✓ *Important Topics* • Evolution of Digital Finance • Traditional Finance vs Digital Finance • Introduction to FinTech • Technological Transformation in Financial Services • Regulatory Frameworks and Compliance in Digital Finance • Role of Big Data in Finance • Big Data and Financial Decision-Making • Big Data in Risk Management • Data Science for Personalization in Financial Services

### *Important Questions*

1. Explain the evolution of digital finance.
2. Differentiate between traditional finance and digital finance.
3. What is FinTech? Explain its importance.
4. Explain the technological transformation in financial services.
5. Explain regulatory frameworks and compliance in digital finance.
6. Discuss the role of Big Data in finance.
7. How does Big Data improve financial decision-making?
8. Explain the role of Big Data in risk management.
9. How is data science used for personalization in financial services?
10. Write short notes on: • FinTech • Big Data in Finance • Digital Transformation • Regulatory Compliance

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## Unit II -- Digital Payment Systems & Digital Banking Transformation

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✓ **Important Topics** • Evolution of Payment Systems • ECS, RTGS, NEFT, IMPS, UPI • Mobile Wallets & Contactless Payments • Attributes of a Well-Functioning Payment System • Role of Banks in Payment Systems • FinTech Startups and Challenger Banks • Peer-to-Peer (P2P) Lending • FinTech Applications in Banking and Insurance • RBI Guidelines and Risks in Digital Payments • Neo-Banks and Future of Digital Banking • Digital Banking Trends and Customer Expectations

### ? **Important Questions**

1. Explain the evolution of digital payment systems.
2. Differentiate between RTGS, NEFT and IMPS.
3. Explain the UPI payment system.
4. Explain mobile wallets and contactless payments.
5. What are the characteristics of a well-functioning payment system?
6. Explain the role of banks in digital payment systems.
7. What are FinTech startups and challenger banks?
8. Explain the concept of Peer-to-Peer (P2P) lending.
9. Discuss the applications of FinTech in banking and insurance.
10. Explain RBI guidelines for digital payment systems.
11. Discuss the risks associated with digital payment systems.
12. Explain neo-banks and the future of digital banking.
13. Write short notes on: • UPI • Mobile Wallets • P2P Lending • Neo-Banks

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## Unit III -- Blockchain, Cryptocurrencies & Decentralized Finance

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✓ **Important Topics** • Fundamentals of Blockchain Technology • Cryptographic Techniques in Blockchain • Smart Contracts • Decentralized Finance (DeFi) • Cryptocurrencies: Bitcoin and Ethereum • Central Bank Digital Currency (CBDC) • Emerging Digital Assets • Blockchain in Payments and Lending • Blockchain in Financial Settlements • Case Study: The Sand Dollar (Bahamas CBDC)

### ? **Important Questions**

1. Explain the fundamentals of blockchain technology.
2. Explain the role of cryptography in blockchain.
3. What are smart contracts? Explain with examples.
4. Explain the concept of Decentralized Finance (DeFi).
5. Discuss the evolution of cryptocurrencies.
6. Explain Bitcoin and Ethereum.
7. What is CBDC? Explain its advantages.
8. Explain the applications of blockchain in payments and lending.
9. Discuss the impact of blockchain on financial settlements.
10. Explain the disruptive potential of blockchain in global finance.
11. Write a note on The Sand Dollar (Bahamas CBDC).
12. Write short notes on: • Blockchain • Smart Contracts • DeFi • Cryptocurrencies

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## Unit IV -- Artificial Intelligence, Machine Learning & Financial Analytics

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✓ *Important Topics* • Artificial Intelligence in Finance • Machine Learning in Finance • Predictive Analytics in Stock Markets • Algorithmic & High-Frequency Trading • Credit Risk Analysis using AI • Automated Decision Making • Data Sourcing, Cleaning & Processing • Data Visualization for Financial Data • Sentiment Analysis • AI-Driven Portfolio Management • Stock Price Prediction Models • Fraud Detection Systems • Credit Score Prediction Models • Digital Lending and Wealth Management

? *Important Questions*

1. Explain the applications of AI in finance.
2. Explain the role of Machine Learning in finance.
3. What is predictive analytics? Explain its applications in stock markets.
4. Explain algorithmic and high-frequency trading.
5. Discuss credit risk analysis using AI.
6. Explain automated decision-making in finance.
7. Explain the process of data sourcing and cleaning.
8. Explain data visualization techniques for financial data.
9. What is sentiment analysis? Explain its applications.
10. Explain AI-driven portfolio management.
11. Explain stock price prediction models.
12. Explain fraud detection systems using AI.
13. Explain credit score prediction models.
14. Discuss the applications of AI in digital lending and wealth management.

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🔥 *Most Important for Exam (High Probability)*

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📖 *Theory* ✓ Evolution of Digital Finance ✓ FinTech and Big Data ✓ RTGS, NEFT, IMPS, UPI ✓ FinTech Startups & P2P Lending ✓ Neo-Banks and Digital Banking Trends ✓ Blockchain Technology ✓ Smart Contracts & DeFi ✓ Cryptocurrencies and CBDCs ✓ AI and Machine Learning in Finance ✓ Sentiment Analysis and Portfolio Management

🎯 *Most Repeated/Scoring Topics*

1. UPI, RTGS, NEFT & IMPS
2. Blockchain & Cryptocurrencies
3. CBDC and The Sand Dollar
4. FinTech & P2P Lending
5. AI Applications in Finance
6. Predictive Analytics & Algorithmic Trading
7. Fraud Detection and Credit Risk Analysis

📚 *Digital Marketing -- Important Topics & Questions (Exam-Oriented)*

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📖 *Unit I -- Introduction to Digital Marketing*

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✓ *Important Topics* • Fundamentals of Digital Marketing • Significance of Digital Marketing • Traditional Marketing vs Digital Marketing • Evolution of Digital Marketing • Digital Marketing Landscape • Key Drivers of Digital Marketing • Digital Users in India • Consumer Decision Journey • Digital Advertising Market in India • Skills Required in Digital Marketing • Digital Marketing Plan

? *Important Questions*

1. Define Digital Marketing and explain its significance.
2. Differentiate between Traditional Marketing and Digital Marketing.
3. Explain the evolution of Digital Marketing.

4. Explain the Digital Marketing Landscape.
5. Discuss the key drivers of Digital Marketing.
6. Explain the growth of digital users in India.
7. Explain the Consumer Decision Journey.
8. Discuss the Digital Advertising Market in India.
9. What skills are required for a Digital Marketer?
10. Explain the steps involved in preparing a Digital Marketing Plan.
11. Write short notes on: • Digital Marketing Landscape • Consumer Decision Journey • Digital Marketing Plan • Digital Advertising in India

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 *Unit II -- Digital Marketing Terminology*

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✓ *Important Topics* • Digital Marketing Terminologies • PPC (Pay-Per-Click) • Social Media Marketing • Google Webmaster Tools • Google Analytics Overview • Email Marketing • Mobile Marketing • Display Advertising • Buying Models in Digital Advertising • Ad Tools and Platforms • Display Advertising Terminology • Types of Display Ads • Ad Formats

? *Important Questions*

1. Explain commonly used terms in Digital Marketing.
2. What is PPC? Explain its advantages.
3. Explain online marketing through social media.
4. Explain the concept of Social Media Marketing.
5. What is Google Webmaster? Explain its features.
6. Explain the importance of Google Analytics.
7. Explain Email Marketing and its advantages.
8. Explain Mobile Marketing.
9. What is Display Advertising?
10. Explain different buying models used in digital advertising.
11. Explain various ad tools and platforms.
12. Discuss different types of display ads and ad formats.
13. Write short notes on: • PPC • Google Analytics • Email Marketing • Display Advertising

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 *Unit III -- Social Media Marketing*

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✓ *Important Topics* • Fundamentals of Social Media Marketing • Significance and Necessity of Social Media Marketing • Facebook Marketing • Facebook for Business • Facebook Insights • Facebook Ad Formats • Setting up a Facebook Advertising Account • Facebook Audience and Audience Types • Designing Facebook Advertising Campaigns • Facebook Avatar • Facebook Apps • Facebook Live • Hashtags

? *Important Questions*

1. Define Social Media Marketing and explain its significance.
2. Why is Social Media Marketing necessary for businesses?
3. Explain Facebook Marketing.
4. Explain the features of Facebook for Business.
5. What are Facebook Insights? Explain their importance.
6. Explain different Facebook Ad formats.
7. Explain the steps to create a Facebook Advertising Account.
8. Explain different types of Facebook audiences.
9. Explain the process of designing a Facebook Advertising Campaign.



10. Write short notes on: • Facebook Avatar • Facebook Live • Hashtags • Facebook Apps

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 *Unit IV -- Search Engine Optimization (SEO)*

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✓ *Important Topics* • Introduction to SEO • How Search Engines Work • SEO Phases • History of SEO • Googlebot (Google Crawler) • Types of SEO Techniques • Keyword Planner Tools • Social Media Reach • Video Creation and Submission • SEO Maintenance and Tactics • Google Search Engine

? *Important Questions*

1. Define SEO and explain its importance.
2. Explain how search engines work.
3. Explain the various phases of SEO.
4. Discuss the history and evolution of SEO.
5. Explain the working of Googlebot.
6. Explain different types of SEO techniques.
7. What is a Keyword Planner Tool? Explain its uses.
8. Explain how social media reach impacts SEO.
9. Explain the role of video creation and submission in SEO.
10. Discuss various SEO tactics and maintenance techniques.
11. Explain the functioning of Google Search Engine.
12. Write short notes on: • Googlebot • Keyword Planner • On-Page SEO • Off-Page SEO

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 *Most Important for Exam (High Probability)*

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 *Theory* ✓ Traditional Marketing vs Digital Marketing ✓ Evolution of Digital Marketing ✓ Consumer Decision Journey ✓ Digital Marketing Plan ✓ PPC and Google Analytics ✓ Email Marketing and Mobile Marketing ✓ Facebook Marketing & Facebook Insights ✓ Search Engine Optimization (SEO) ✓ Googlebot and Search Engine Working ✓ Keyword Planner Tools

★ *Very Important Topics* ✓ Facebook Advertising Campaign ✓ Display Advertising and Ad Formats ✓ Social Media Marketing ✓ Types of SEO Techniques ✓ SEO Tactics and Maintenance

🎯 *Most Repeated/Scoring Topics*

1. Traditional Marketing vs Digital Marketing
2. Consumer Decision Journey
3. PPC and Google Analytics
4. Facebook Marketing & Ad Campaigns
5. Search Engine Optimization (SEO)
6. Googlebot and Search Engine Working
7. On-Page SEO and Off-Page SEO
8. Keyword Planner Tools

 *Digital Business -- Important Topics & Questions (Exam-Oriented)*

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 *Unit I -- Foundations of Digital Business*

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✓ *Important Topics* • Digital Economy and Digital Society • Digital Transformation and its Impact • Digital Enterprise: Strategies and Characteristics • Online and Virtual Communities • Definition, Scope and Evolution of Digital Business • Emerging Platforms:

- AI-Driven Marketplaces
- Blockchain-Based Systems
- Metaverse Commerce

- E-Business Models:
  - B2B
  - B2C
  - C2C
  - D2C
  - Hybrid Models
- Physical and Digital Marketplaces • Web 3.0 and Decentralized Platforms • Drivers:
  - Scalability
  - Data Analytics
  - Automation
  - Artificial Intelligence
- Benefits:
  - Global Reach
  - Personalization
  - Efficiency
- Limitations:
  - Cybersecurity Risks
  - Digital Divide
- Impact on Businesses, Customers, Governments and Society

### *Important Questions*

1. Define Digital Business and explain its scope and evolution.
2. Explain the concept of Digital Economy and Digital Society.
3. Explain Digital Transformation and its impact.
4. What is a Digital Enterprise? Explain its characteristics.
5. Explain various E-Business Models.
6. Differentiate between B2B, B2C, C2C and D2C models.
7. Explain AI-driven marketplaces and Metaverse Commerce.
8. What is Web 3.0? Explain decentralized platforms.
9. Explain the drivers and benefits of Digital Business.
10. Discuss the limitations of Digital Business.
11. Explain the impact of Digital Business on society and government.

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## *Unit II -- Mobile, Social and IoT-Driven Commerce*

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 *Important Topics* • Mobile Commerce (M-Commerce) • Evolution and Applications of Mobile Commerce • Impact of 5G on Mobile Commerce • Mobile Marketing • Location-Based Advertising • In-App Purchases • Mobile Wallets • Social Commerce • Social Media Marketplaces • Influencer Economy • Live-Streaming Commerce • Social Business Networks • Enterprise Collaboration Tools • Social Media Analytics • Customer Engagement and Brand Loyalty • Privacy Concerns and Platform Dependency • Internet of Things (IoT) • IoT in Retail, Supply Chain, Smart Cities and Healthcare • Connected Devices • Predictive Maintenance • Edge Computing • Wearables

### *Important Questions*

1. Explain the evolution and applications of Mobile Commerce.
2. Discuss the impact of 5G on Mobile Commerce.

3. Explain Mobile Marketing and its applications.
4. Explain location-based advertising and in-app purchases.
5. What is Social Commerce? Explain its importance.
6. Explain the Influencer Economy and Live-Streaming Commerce.
7. Explain Social Business Networks and Enterprise Collaboration Tools.
8. Explain Social Media Analytics.
9. Discuss the benefits and limitations of Social Commerce.
10. Explain the concept of IoT.
11. Explain IoT applications in retail and healthcare.
12. Explain Predictive Maintenance and Edge Computing.
13. Write short notes on: • Mobile Wallets • Wearables • Connected Devices • Smart Cities

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 *Unit III -- Digital Business Ecosystem*

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 **Important Topics** • Digital Commerce Mechanisms • Online Purchasing and E-Marketplaces • Digital Twins • Multi-Sided Platforms • Platform Economy • Disintermediation and Reintermediation • Customer Mechanisms:

- Webstores
- Malls
- Portals
- Mobile Apps
- Voice Commerce
- Intermediaries:
  - Aggregators
  - Curators
- Merchant Solutions:
  - Electronic Catalogs
  - Search Engines
  - Recommendation Systems
  - Shopping Carts
- E-Auctions and Algorithmic Pricing • Supply Chain Technologies:
  - Blockchain
  - 3D Printing
  - Just-in-Time Delivery
- Digital Payments:
  - Mobile Payments
  - Digital Wallets
  - Cryptocurrencies
  - CBDCs
- Security, Privacy and Ethical Issues

 **Important Questions**

1. Explain Digital Commerce Mechanisms.
2. Explain E-Marketplaces and Online Purchasing.
3. What is a Platform Economy? Explain with examples.
4. Differentiate between Disintermediation and Reintermediation.

5. Explain Digital Twins and their applications.
6. Explain Webstores, Portals and Voice Commerce.
7. Explain the role of Aggregators and Curators.
8. Explain Recommendation Systems and Shopping Carts.
9. What are E-Auctions? Explain algorithmic pricing.
10. Explain the role of Blockchain in supply chains.
11. Explain Digital Payments and Digital Wallets.
12. Discuss security and privacy issues in Digital Business.

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#### Unit IV -- Digital Business Applications

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 **Important Topics** • Electronic Retailing (B2C) • Omnichannel Retailing • AI-Driven Personalization • Social Shopping • User-Generated Content • Social Proof • FinTech • E-Banking and Mobile Banking • Neo-Banks • InsurTech • Open Banking • RegTech • Digital Government • Smart Cities • E-Governance • Data-Driven Policy Making • Digital Public Services • E-Learning and EdTech • Online Platforms and Corporate Training • Gamification • VR/AR in Education • Digital Content:

- Streaming Services
- Podcasts
- Content Monetization

#### **Important Questions**

1. Explain Electronic Retailing and Omnichannel Retailing.
2. Explain AI-driven personalization in Digital Business.
3. What is Social Shopping? Explain its benefits.
4. Explain User-Generated Content and Social Proof.
5. Explain the concept of FinTech.
6. Differentiate between E-Banking and Mobile Banking.
7. Explain Neo-Banks, InsurTech and RegTech.
8. Explain Open Banking and its advantages.
9. Discuss the concept of Digital Government and Smart Cities.
10. Explain E-Governance and Digital Public Services.
11. Explain the role of E-Learning and EdTech.
12. Explain Gamification and VR/AR in Education.
13. Discuss content monetization and streaming services.

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#### Most Important for Exam (High Probability)

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 **Theory** ✓ Digital Transformation and Digital Enterprise ✓ E-Business Models (B2B, B2C, C2C, D2C) ✓ Web 3.0 and Metaverse Commerce ✓ Mobile Commerce and Social Commerce ✓ Internet of Things (IoT) ✓ Platform Economy ✓ Disintermediation and Reintermediation ✓ Digital Payments and CBDCs ✓ FinTech and Neo-Banks ✓ E-Governance and Smart Cities ✓ E-Learning and EdTech

 **Very Important Topics** ✓ AI-Driven Marketplaces ✓ Influencer Economy ✓ Digital Twins ✓ Recommendation Systems ✓ Blockchain in Supply Chains ✓ Open Banking and RegTech ✓ VR/AR in Education

#### **Most Repeated/Scoring Topics**

1. Digital Transformation
2. E-Business Models
3. Mobile Commerce & Social Commerce

4. Internet of Things (IoT)
5. Platform Economy
6. Digital Payments & Digital Wallets
7. FinTech & Neo-Banks
8. Smart Cities & E-Governance
9. EdTech and Gamification

 *Digital Business and Technology -- Important Topics & Questions (Exam-Oriented)*

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 *Unit I -- Introduction to Digital Business*

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 *Important Topics* • Introduction to Digitization • Impact of Digitization on Business • Social Media Marketing • Digital Business Models • Concept of Digital Marketing and its Impact • Digital Strategy • Innovation in Digital Business

 *Important Questions*

1. Define Digitization and explain its importance.
2. Explain the impact of digitization on business.
3. Explain the concept of Social Media Marketing.
4. What is a Digital Business Model?
5. Explain the concept and impact of Digital Marketing.
6. What is Digital Strategy? Explain its importance.
7. Explain the role of innovation in Digital Business.
8. Differentiate between Traditional Business and Digital Business.
9. Write short notes on: • Digitization • Digital Marketing • Social Media Marketing • Digital Strategy

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 *Unit II -- Digital Business Model*

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 *Important Topics* • Introduction to Digital Business Model Innovation • Key Drivers of Digital Business Model Reinvention • Types of Digital Business Models • Reinvented Business Organizations • Subscription-Based Streaming Models • Original Content Production • AI-Based Personalized Recommendations

 *Important Questions*

1. Explain the concept of Digital Business Model Innovation.
2. Discuss the key drivers of Digital Business Model Reinvention.
3. Explain various types of Digital Business Models.
4. Explain how organizations reinvent their business models.
5. Explain the subscription-based streaming business model.
6. How is AI used for personalized recommendations?
7. Explain the importance of original content production in digital businesses.
8. Discuss a case study of a reinvented business organization.
9. Write short notes on: • Subscription-Based Model • Personalized Recommendations • Digital Business Model Innovation • AI in Business Models

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 *Unit III -- Business Automation and Cyber Security*

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 *Important Topics* • Introduction to Automation in Digital Business • Role of Automation • Automation Technologies • Automation Implementation and Integration • Impact of Automation on Digital Business • Introduction to Cyber Security • Cyber Security Measures • Cyber Security Best Practices

### *Important Questions*

1. Explain the concept of Automation in Digital Business.
2. Discuss the role of Automation in Digital Business.
3. Explain various Automation Technologies.
4. Explain the implementation and integration of Automation.
5. Discuss the impact of Automation on businesses.
6. Define Cyber Security and explain its importance.
7. Explain various Cyber Security Measures.
8. Discuss Cyber Security Best Practices.
9. Explain the challenges of Cyber Security in Digital Business.
10. Write short notes on: • Automation Technologies • Cyber Security Measures • Data Protection • Best Practices in Cyber Security

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### *Unit IV -- Emerging Technology and Entrepreneurship*

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### *Important Topics* • Role of Technology in Modern Startups • Digital Marketing Fundamentals:

- SEO
- Social Media Marketing
- Email Marketing
- E-Commerce Platforms:
  - Shopify
  - WooCommerce
- Introduction to:
  - Artificial Intelligence (AI)
  - Internet of Things (IoT)
  - Blockchain
- Automation of Tasks • Data Analysis using AI • Personalized User Experiences • Smart Product Development

### *Important Questions*

1. Explain the role of technology in modern startups.
2. Explain the fundamentals of Digital Marketing.
3. Explain Search Engine Optimization (SEO).
4. Explain Social Media Marketing and Email Marketing.
5. Explain the features of Shopify and WooCommerce.
6. Discuss the applications of AI in startups.
7. Explain the role of IoT in entrepreneurship.
8. Explain the applications of Blockchain in startups.
9. How can AI automate tasks and analyze data?
10. Explain personalized user experiences using AI.
11. Explain the development of smart products using emerging technologies.
12. Write short notes on: • AI in Startups • IoT Applications • Blockchain Technology • E-Commerce Platforms

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### *Most Important for Exam (High Probability)*

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 *Theory* ✓ Digitization and its Impact on Business ✓ Digital Business Models ✓ Digital Strategy and Innovation ✓ Digital Business Model Innovation ✓ Subscription-Based Streaming Model ✓ Automation in Digital Business ✓ Cyber Security Measures and Best Practices ✓ Role of Technology in Startups ✓ AI, IoT and Blockchain in Startups ✓ SEO, Social Media and Email Marketing

★ *Very Important Topics* ✓ AI-Based Personalized Recommendations ✓ Automation Technologies ✓ E-Commerce Platforms (Shopify, WooCommerce) ✓ Smart Product Development ✓ Data Analysis using AI

 *Most Repeated/Scoring Topics*

1. Digitization and Digital Business
2. Digital Business Models
3. Subscription-Based Streaming Case Study
4. Automation in Digital Business
5. Cyber Security Measures
6. SEO and Digital Marketing Fundamentals
7. AI, IoT and Blockchain Applications
8. Technology in Modern Startups
9. Personalized User Experiences using AI

 *Personal Financial Management -- Important Topics & Questions (Exam-Oriented)*

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 *Unit I -- Introduction to Personal Finance and Budgeting*

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✓ *Important Topics* • Importance and Scope of Personal Financial Management • Financial Goal Setting:

- Short-Term Goals
- Medium-Term Goals
- Long-Term Goals

• Personal Income and Expenditure Planning • Budgeting Techniques and Tracking Tools • Manual and Digital Budgeting Tools • Emergency Fund Planning • Financial Discipline and Behavioral Aspects of Money

? *Important Questions*

1. Define Personal Financial Management and explain its importance.
2. Explain the scope of Personal Financial Management.
3. Explain short-term, medium-term and long-term financial goals.
4. Explain the process of personal income and expenditure planning.
5. Discuss various budgeting techniques.
6. Explain manual and digital budgeting tools.
7. What is an emergency fund? Explain its importance.
8. Explain the importance of financial discipline.
9. Discuss the behavioral aspects of money management.
10. Write short notes on: • Financial Goal Setting • Budgeting Techniques • Emergency Fund • Financial Discipline

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 *Unit II -- Banking, Credit and Digital Finance*

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✓ *Important Topics* • Basics of Banking • Types of Bank Accounts • Bank Statements and Interest • Digital Banking Tools:

- UPI
  - NEFT
  - RTGS
  - Mobile Banking
  - E-Wallets
- Credit Cards and Debit Cards • Responsible Use of Credit Cards • Student Loans, Personal Loans and Vehicle Loans • Credit Score and its Importance • Factors Affecting Credit Score • Digital Security • Phishing and Fraud Prevention • Cyber Hygiene

### *Important Questions*

1. Explain the basics of banking.
2. Explain different types of bank accounts.
3. Explain the importance of bank statements.
4. Differentiate between UPI, NEFT and RTGS.
5. Explain mobile banking and e-wallets.
6. Differentiate between credit cards and debit cards.
7. Explain the responsible use of credit cards.
8. Explain student, personal and vehicle loans.
9. What is a credit score? Explain its importance.
10. Explain the factors affecting credit score.
11. What is phishing? Explain fraud prevention techniques.
12. Explain cyber hygiene and digital security practices.
13. Write short notes on: • UPI • Credit Score • Mobile Banking • Cyber Hygiene

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### *Unit III -- Saving and Investment Options*

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#### *Important Topics* • Importance of Saving • Saving Schemes:

- Fixed Deposit (FD)
  - Recurring Deposit (RD)
  - Public Provident Fund (PPF)
- Introduction to Investment • Risk vs Return • Mutual Funds and SIPs • Compounding • Time Value of Money • Investment Avenues:
- Gold
  - Real Estate
  - Stock Market
- Financial Planning Apps and Tools

### *Important Questions*

1. Explain the importance of saving.
2. Explain various saving schemes such as FD, RD and PPF.
3. Explain the concept of investment.
4. Differentiate between risk and return.
5. Explain Mutual Funds and SIPs.
6. Explain the concept of compounding.
7. Explain the Time Value of Money.
8. Explain various investment avenues.
9. Compare gold, real estate and stock market investments.



10. Explain the role of financial planning apps and tools.
11. Write short notes on: • SIP • Mutual Funds • Compounding • Time Value of Money

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 *Unit IV -- Insurance, Tax Basics and Retirement Planning*

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 **Important Topics** • Concept of Insurance • Types of Insurance:

- Life Insurance
- Health Insurance
- General Insurance
- Nomination and Claims • Basics of Income Tax • Income Tax Slabs • PAN • Tax-Saving Instruments
- Retirement Planning • EPF (Employee Provident Fund) • NPS (National Pension System) • Pension Schemes • Common Financial Frauds and Safety Tips • Ethical Financial Behavior • Long-Term Wealth Planning

 **Important Questions**

1. Define Insurance and explain its importance.
2. Explain various types of insurance.
3. Explain the process of nomination and claim settlement.
4. Explain the basics of Income Tax.
5. Explain Income Tax Slabs and PAN.
6. Explain various tax-saving instruments.
7. Explain the importance of retirement planning.
8. Explain EPF and NPS.
9. Explain various pension schemes.
10. Discuss common financial frauds and safety tips.
11. Explain ethical financial behavior.
12. Explain long-term wealth planning.
13. Write short notes on: • Life Insurance • Health Insurance • EPF and NPS • Financial Frauds

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 **Most Important for Exam (High Probability)**

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 **Theory** ✓ Financial Goal Setting ✓ Budgeting Techniques ✓ Emergency Fund Planning ✓ UPI, NEFT and RTGS ✓ Credit Score and Factors Affecting It ✓ Mutual Funds and SIPs ✓ Compounding and Time Value of Money ✓ Insurance and Types of Insurance ✓ EPF, NPS and Retirement Planning ✓ Financial Frauds and Cyber Security

 **Very Important Topics** ✓ Risk vs Return ✓ Saving Schemes (FD, RD, PPF) ✓ Credit and Debit Cards ✓ Tax-Saving Instruments ✓ Long-Term Wealth Planning

 **Most Repeated/Scoring Topics**

1. Budgeting and Financial Goal Setting
2. Digital Banking Tools (UPI, NEFT, RTGS)
3. Credit Score and Loans
4. Mutual Funds and SIPs
5. Compounding and Time Value of Money
6. Insurance and Claims
7. Income Tax Basics and PAN
8. EPF, NPS and Pension Schemes
9. Financial Frauds and Safety Tips

✓ *Important Topics* • Introduction to Different Beverages • Concept of Wine • Health Benefits of Wine • History and Evolution of Winemaking • Major Wine-Producing Regions • Wine Classifications and Types:

- Red Wine
  - White Wine
  - Rosé Wine
  - Sparkling Wine
  - Fortified Wine
- Basic Winemaking Process • Important Wine Terminologies • Importance of Grapes in Winemaking  
• Grape Varieties and Their Impact on Wine Quality • Global Wine Industry Overview

? *Important Questions*

1. Define wine and explain its characteristics.
2. Explain the history and evolution of winemaking.
3. Explain the health benefits of wine.
4. Discuss the major wine-producing regions of the world.
5. Classify wines and explain different types of wine.
6. Explain the basic winemaking process.
7. Explain important wine terminologies.
8. Discuss the importance of grapes in winemaking.
9. Explain how grape varieties affect wine quality.
10. Give an overview of the global wine industry.
11. Write short notes on: • Red and White Wine • Sparkling Wine • Fortified Wine • Major Wine-Producing Regions

✓ *Important Topics* • Wine Making Process-I (Pre-Fermentation)

- Harvesting of Grapes
  - Crushing
  - Preparation and Extraction of Must
  - Maceration
- Wine Making Process-II (Fermentation)
- Concept of Fermentation
  - Types of Fermenters
  - Parts of Fermenter
  - Parameters Affecting Fermentation
- Wine Making Process-III (Post-Fermentation)
- Flavour Enhancement
  - Aging of Wine
  - Barrel Aging
  - Bottle Maturation
  - Quality Control

- Bottling
- Cellar System

### *Important Questions*

1. Explain the pre-fermentation process in winemaking.
2. Explain harvesting and crushing of grapes.
3. What is must? Explain its preparation and extraction.
4. Explain the process of maceration.
5. Define fermentation and explain its importance.
6. Explain different types of fermenters.
7. Explain the parameters affecting fermentation.
8. Explain flavour enhancement and aging of wine.
9. Discuss barrel aging and bottle maturation.
10. Explain quality control in wine production.
11. Explain bottling and cellar systems.
12. Draw and explain the complete winemaking process.

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### *Unit III -- Sensory Analysis and Quality Control*

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#### *Important Topics* • Sensory Evaluation and Terminology • Human Sensory Organs and Perception

##### • Sensory Attributes of Wine:

- Appearance
- Aroma
- Mouthfeel

##### • Glassware Selection and Preparation for Wine Tasting • Storage and Handling of Wine • Types of Sensory Tests:

- Discrimination Test
- Descriptive Test
- Hedonic Test

##### • Common Off-Flavors and Wine Defects

### *Important Questions*

1. Define sensory evaluation and explain its importance.
2. Explain human sensory organs and perception in wine tasting.
3. Explain the sensory attributes of wine.
4. Explain the importance of aroma and mouthfeel in wine evaluation.
5. Discuss glassware selection for wine tasting.
6. Explain proper storage and handling of wine.
7. Explain different types of sensory tests.
8. Differentiate between discrimination, descriptive and hedonic tests.
9. Explain common off-flavors and defects in wine.
10. Write short notes on: • Wine Aroma • Wine Appearance • Mouthfeel • Wine Defects

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### *Unit IV -- Wine Economics and Trade*

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#### *Important Topics* • Role of Wine in Hospitality and Tourism • Wine Tourism Destinations • Wine Business Models • Global Wine Market Trends • Career Opportunities in the Wine Industry

### *Important Questions*

1. Explain the role of wine in hospitality and tourism.
2. Discuss the importance of wine tourism.
3. Explain various wine business models.
4. Discuss global wine market trends.
5. Explain the economic importance of the wine industry.
6. Discuss career opportunities in the wine industry.
7. Write short notes on: • Wine Tourism • Global Wine Market • Hospitality Industry • Career Opportunities in Wine Industry

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 *Most Important for Exam (High Probability)*

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 *Theory* ✓ History and Evolution of Winemaking ✓ Wine Classification and Types ✓ Grape Varieties and Wine Quality ✓ Complete Winemaking Process ✓ Fermentation and Types of Fermenters ✓ Barrel Aging and Bottle Maturation ✓ Sensory Evaluation and Wine Tasting ✓ Storage and Handling of Wine ✓ Wine Tourism and Business Models ✓ Global Wine Market Trends

★ *Very Important Topics* ✓ Major Wine-Producing Regions ✓ Parameters Affecting Fermentation ✓ Quality Control in Wine Production ✓ Types of Sensory Tests ✓ Common Wine Defects and Off-Flavors

 *Most Repeated/Scoring Topics*

1. Types of Wine
2. Complete Winemaking Process
3. Fermentation Process
4. Sensory Attributes of Wine
5. Wine Storage and Handling
6. Wine Tourism
7. Global Wine Industry Overview
8. Career Opportunities in Wine Industry

 *Dairy Technology -- Important Topics & Questions (Exam-Oriented)*

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 *Unit I -- Livestock Diversity and its Management*

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✓ *Important Topics* • Livestock Diversity in India and its Importance • Role of Livestock in Agriculture • Selection of Elite Animals • Characteristics of an Ideal Dairy Farm • Management of Milking Animals • Animal Health Management and its Impact on Milk Quality • Milking Systems • Hygienic Milk Production

❓ *Important Questions*

1. Explain the importance of livestock diversity in India.
2. Discuss the role of livestock in agriculture.
3. Explain the selection criteria for elite dairy animals.
4. Explain the characteristics of an ideal dairy farm.
5. Explain the management of milking animals.
6. Discuss animal health management and its impact on milk quality.
7. Explain different milking systems.
8. Explain the importance of hygienic milk production.
9. Write short notes on: • Elite Animals • Dairy Farm Management • Animal Health Management • Hygienic Milk Production

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 *Unit II -- Milk Process Technology*

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✓ *Important Topics* • Nutritional Importance of Milk and its Constituents • Collection Methods of Raw Milk • Sampling Methods of Raw Milk • Quality Assessment of Raw Milk • Milk Processing:

- Cooling
- Separation
- Standardization
- Homogenization
- Pasteurization
- Types of Pasteurization • Types of Milk:
  - Pasteurized Milk
  - Standardized Milk
  - Toned Milk
  - Double Toned Milk
  - Flavoured Milk
- Production of Milk Products:
  - Yogurt
  - Butter
  - Buttermilk
  - Paneer
- Dairy Plant Hygiene and Sanitation • Disposal of Dairy Waste

❓ *Important Questions*

1. Explain the nutritional importance of milk.
2. Explain various methods of raw milk collection.
3. Explain the sampling methods of raw milk.
4. Discuss the quality assessment of raw milk.
5. Explain the processing of milk and its significance.
6. Explain cooling, separation and standardization of milk.
7. Explain homogenization and pasteurization.
8. Explain different types of pasteurization.
9. Differentiate between toned and double-toned milk.
10. Explain the production of yogurt, butter and paneer.
11. Explain dairy plant hygiene and sanitation.
12. Discuss methods of dairy waste disposal.

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 *Unit III -- Quality Assurance of Milk*

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✓ *Important Topics* • Spoilage of Milk and Milk Products by Microorganisms • Control of Milk Spoilage • Packaging Materials for Dairy Products • Importance of Packaging in Preservation • Quality Assurance of Dairy Products • ISO 9001:2000 • HACCP (Hazard Analysis and Critical Control Points) • Detection of Milk Adulteration • Case Study:

- Milk Packaging Methods
- Types of Paper Used for Packaging
- Quality Checks

❓ *Important Questions*

1. Explain the spoilage of milk by microorganisms.
2. Discuss methods for controlling milk spoilage.

3. Explain different packaging materials used for milk.
4. Why is packaging important in milk preservation?
5. Explain ISO 9001:2000 in dairy industries.
6. Explain the HACCP food safety system.
7. Explain quality assurance of dairy products.
8. Explain methods for detecting milk adulteration.
9. Discuss the case study on milk packaging and quality checks.
10. Write short notes on: • HACCP • ISO 9001:2000 • Milk Adulteration • Packaging Materials

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 *Unit IV -- Case Studies in Dairy Technology*

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 **Important Topics** • Butter Technology and Business Model • Cheese Technology and Business Model • Paneer Technology and Business Model • Ice Creams and Frozen Desserts • Packaging of Dairy Products • Refrigeration and Air Conditioning in Dairy Industry • Human Nutritional Characteristics

 **Important Questions**

1. Explain butter technology and its business model.
2. Explain cheese technology and its business model.
3. Explain paneer technology and its business model.
4. Explain the production of ice creams and frozen desserts.
5. Discuss packaging techniques for dairy products.
6. Explain the role of refrigeration and air conditioning in dairy industries.
7. Explain the human nutritional characteristics of dairy products.
8. Write short notes on: • Butter Technology • Cheese Technology • Frozen Desserts • Refrigeration in Dairy Industry

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 **Most Important for Exam (High Probability)**

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 **Theory** ✓ Livestock Diversity in India ✓ Characteristics of an Ideal Dairy Farm ✓ Animal Health Management ✓ Milk Processing and Pasteurization ✓ Types of Milk and their Production ✓ Production of Yogurt, Butter and Paneer ✓ Milk Spoilage and its Control ✓ Packaging Materials and Preservation ✓ HACCP and ISO 9001:2000 ✓ Refrigeration and Packaging of Dairy Products

 **Very Important Topics** ✓ Quality Assessment of Raw Milk ✓ Detection of Milk Adulteration ✓ Dairy Plant Hygiene and Sanitation ✓ Butter, Cheese and Paneer Technology ✓ Human Nutritional Characteristics

 **Most Repeated/Scoring Topics**

1. Characteristics of Ideal Dairy Farm
2. Pasteurization and Homogenization
3. Types of Milk and Milk Products
4. Quality Assurance of Dairy Products
5. HACCP and ISO 9001:2000
6. Milk Adulteration Detection
7. Packaging of Milk and Dairy Products
8. Refrigeration in Dairy Industry

 **Supply Chain Management -- Important Topics & Questions (Exam-Oriented)**

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 *Unit I -- Supply Chain Structure & Flow*

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 **Important Topics** • Shift from Enterprise to Network • Structure of Supply Chain (SC) • Push-Based Supply Chain • Pull-Based Supply Chain • Trade-off between Push and Pull Strategy •

Identifying Appropriate Push-Pull Strategy • Commodity and Cost-Centric Supply Chain • Agile Supply Chain • Forward and Reverse Supply Chain • Flows in Supply Chain:

- Product Flow
- Service Flow
- Information Flow
- Funds Flow
- Demand Flow
- Forecast Flow

• Upstream and Downstream Supply Chain

 *Important Questions*

1. Explain the structure of a Supply Chain.
2. Explain the shift from enterprise to network.
3. Explain Push-Based and Pull-Based Supply Chains.
4. Differentiate between Push and Pull strategies.
5. Explain the trade-off between Push and Pull strategies.
6. How do you identify an appropriate Push-Pull strategy?
7. Explain Commodity and Cost-Centric Supply Chain.
8. Explain Agile Supply Chain.
9. Differentiate between Forward and Reverse Supply Chain.
10. Explain various flows in Supply Chain Management.
11. Differentiate between Upstream and Downstream Supply Chains.
12. Write short notes on: • Agile Supply Chain • Reverse Supply Chain • Information Flow • Forecast Flow

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 *Unit II -- Total Supply Chain Management*

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 *Important Topics* • Business Landscape and Driving Forces • Shift from Operations to Services • Impact of Globalization • Impact of Technological Revolution • Shift from Linear Supply Chain to Collaborative Networks • Power Shifts in Supply Chain • Flexibility of Partnerships • Core Competencies • Growth in Outsourcing

 *Important Questions*

1. Explain the changing business landscape in Supply Chain Management.
2. Explain the shift from operations to services.
3. Discuss the impact of globalization on Supply Chain Management.
4. Explain the impact of technological revolution on Supply Chains.
5. Explain the shift from linear supply chains to collaborative networks.
6. Discuss power shifts in Supply Chain Management.
7. Why is flexibility important in supply chain partnerships?
8. Explain the concept of core competencies.
9. Explain the role of outsourcing in Supply Chain Management.
10. Write short notes on: • Globalization • Collaborative Networks • Core Competencies • Outsourcing

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 *Unit III -- Supply Chain Management Building Blocks*

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 *Important Topics* • Customer Focus and Demand Management • Resources and Capacity Management • Procurement and Supplier Focus • Inventory Management • Operations Management •

## Distribution Management in SCM

### ? Important Questions

1. Explain customer focus and demand management in SCM.
2. Explain resources and capacity management.
3. Discuss procurement and supplier focus.
4. Explain inventory management and its importance.
5. Explain operations management in SCM.
6. Explain distribution management in Supply Chain Management.
7. Discuss the building blocks of Supply Chain Management.
8. Write short notes on: • Demand Management • Procurement • Inventory Management • Distribution Management

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## Unit IV -- Customer Value

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✓ Important Topics • Empowered Consumer • Customer-Focused Marketing and Supply Chain Service Outputs • Customer Service:

- Availability
- Operational Performance
- Reliability
- Customer Satisfaction:
  - Customer Expectations
  - Enhancing Customer Satisfaction
  - Limitations of Customer Satisfaction
- Customer Success:
  - Achieving Customer Success
  - Value-Added Services
  - Customer Value Requirement Mapping

### ? Important Questions

1. Explain the concept of an empowered consumer.
2. Explain customer-focused marketing and service outputs.
3. Explain the elements of customer service.
4. Discuss availability, operational performance and reliability.
5. What is customer satisfaction? Explain its importance.
6. Explain customer expectations and methods to enhance satisfaction.
7. Discuss the limitations of customer satisfaction.
8. Explain the concept of customer success.
9. Explain value-added services in Supply Chain Management.
10. Explain customer value requirement mapping.
11. Write short notes on: • Customer Satisfaction • Customer Success • Reliability • Value-Added Services

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## Most Important for Exam (High Probability)

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 Theory ✓ Push-Based and Pull-Based Supply Chain ✓ Trade-off between Push and Pull Strategy ✓ Agile, Forward and Reverse Supply Chain ✓ Supply Chain Flows (Product, Information, Funds, Demand) ✓ Impact of Globalization and Technology ✓ Collaborative Networks and Outsourcing ✓



Inventory Management and Distribution Management ✓ Customer Satisfaction and Customer Success ✓ Customer Value Requirement Mapping

★ *Very Important Topics* ✓ Structure of Supply Chain ✓ Upstream and Downstream Supply Chain ✓ Core Competencies ✓ Procurement and Supplier Focus ✓ Customer Service Outputs

🎯 *Most Repeated/Scoring Topics*

1. Push vs Pull Supply Chain
2. Agile and Reverse Supply Chain
3. Globalization and Technological Revolution
4. Building Blocks of SCM
5. Inventory Management
6. Customer Service and Satisfaction
7. Customer Success
8. Value-Added Services and Customer Value Mapping

📖 *Digital Manufacturing -- Important Topics & Questions (Exam-Oriented)*

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📖 *Unit I -- 3D Digital Factories*

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✓ *Important Topics* • The Promise of 3D Digital Factories • Digital Design and New Workflows • 3D Additive Printing (3D Printing) • Integration of Operational Technology (OT) and Information Technology (IT) • Conceptual Design of a 3D Digital Factory

? *Important Questions*

1. Define a 3D Digital Factory and explain its importance.
2. Explain the concept and features of a 3D Digital Factory.
3. Explain digital design and new workflows in manufacturing.
4. What is 3D Additive Printing? Explain its advantages.
5. Explain the integration of Operational Technology and Information Technology.
6. Explain the conceptual design of a 3D Digital Factory.
7. Discuss the benefits and challenges of Digital Factories.
8. Write short notes on: • 3D Printing • Digital Design • OT and IT Integration • Digital Factory

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📖 *Unit II -- Digital Twin Technology*

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✓ *Important Topics* • Production Planning and Scheduling in Smart Factories • Concept of Digital Twin • Cyber-Physical Integration • Implementing Digital Twin Technology • Industrial Case Studies • Smart Production Resource Allocation

? *Important Questions*

1. Explain production planning and scheduling in a smart factory.
2. Define Digital Twin and explain its importance.
3. Explain Cyber-Physical Integration.
4. Explain the implementation of Digital Twin technology.
5. Discuss the applications of Digital Twin in industries.
6. Explain smart production resource allocation.
7. Explain the advantages of Digital Twin technology.
8. Discuss industrial case studies based on Digital Twin.
9. Write short notes on: • Smart Factory • Cyber-Physical Systems • Digital Twin Applications • Resource Allocation

✓ *Important Topics* • Knowledge Discovery and Extraction • Knowledge Representation and Reasoning • Construction of Industrial Knowledge Graph • Knowledge Graph-Enabled Knowledge Evolvement • Industrial Case Studies

? *Important Questions*

1. Explain knowledge discovery and extraction.
2. Explain knowledge representation and reasoning.
3. What is an Industrial Knowledge Graph?
4. Explain the construction of an Industrial Knowledge Graph.
5. Explain Knowledge Graph-enabled knowledge evolvement.
6. Discuss the applications of knowledge management in industries.
7. Explain the importance of Engineering Knowledge Management.
8. Discuss industrial case studies related to knowledge management.
9. Write short notes on: • Knowledge Discovery • Knowledge Representation • Industrial Knowledge Graph • Knowledge Evolvement

*Aspects*

✓ *Important Topics* • Business Models for the New Enterprise • Supply Chain Strategies • Additive Manufacturing and Supply Chain Resiliency • Design Customization and Optimization • Risks and Threats in Distributed Digitized Manufacturing • Modern Security for Digital Manufacturing

? *Important Questions*

1. Explain business models for the new enterprise.
2. Discuss various supply chain strategies in digital manufacturing.
3. Explain additive manufacturing and supply chain resiliency.
4. Explain design customization and optimization.
5. Discuss risks and threats in distributed digitized manufacturing.
6. Explain modern security measures for digital manufacturing.
7. Discuss the challenges of digital manufacturing.
8. Explain the importance of cybersecurity in digital manufacturing.
9. Write short notes on: • Supply Chain Resiliency • Additive Manufacturing • Design Optimization • Security in Digital Manufacturing

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 *Most Important for Exam (High Probability)*

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 *Theory* ✓ 3D Digital Factory and its Architecture ✓ 3D Additive Printing ✓ Integration of OT and IT ✓ Digital Twin Technology ✓ Cyber-Physical Systems ✓ Smart Production Resource Allocation ✓ Engineering Knowledge Management ✓ Industrial Knowledge Graph ✓ Supply Chain Strategies ✓ Security in Digital Manufacturing

★ *Very Important Topics* ✓ Conceptual Design of a 3D Digital Factory ✓ Production Planning in Smart Factories ✓ Knowledge Representation and Reasoning ✓ Additive Manufacturing and Supply Chain Resiliency ✓ Risks and Threats in Digitized Manufacturing

 *Most Repeated/Scoring Topics*

1. 3D Digital Factory
2. 3D Additive Printing
3. Digital Twin Technology

4. Cyber-Physical Integration
5. Industrial Knowledge Graph
6. Supply Chain Strategies
7. Additive Manufacturing
8. Security Aspects of Digital Manufacturing

### *Maharashtra Land Laws -- Important Topics & Questions (Exam-Oriented)*

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#### *Unit I -- Maharashtra Land Revenue Code, 1966*

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 **Important Topics** • Historical Background of Land Revenue System • Object, Application and Definitions under the Code • Revenue Officers -- Powers and Duties • Vesting and Extinction of Rights in Land • Assignment of Land for Special Purposes • Rights relating to Trees and Forests • Grant and Use of Land • Encroachment on Land • Relinquishment and Surrender of Land • Land Revenue -- Liability and Assessment (Sections 64-78) • Revenue Surveys and Procedure (Sections 79-88) • Survey Numbers, Partitions and Sub-divisions • Boundary and Boundary Marks (Sections 132-146) • Procedure of Revenue Officers • Appeals, Revision and Review • Maharashtra Revenue Tribunal (MRT) • Record of Rights (Sections 147-159) • Rights in Unoccupied Lands (Sections 160-167) • Realization and Recovery of Land Revenue (Sections 168-184)

#### **Important Questions**

1. Explain the historical background of the Maharashtra Land Revenue Code, 1966.
2. Explain the powers and duties of Revenue Officers.
3. Explain vesting and extinction of rights in land.
4. Explain grant and use of land under the Code.
5. Explain encroachment, relinquishment and surrender of land.
6. Explain liability and assessment of land revenue.
7. Explain the procedure of revenue surveys.
8. Explain boundary and boundary marks.
9. Explain the Record of Rights.
10. Explain rights in unoccupied lands.
11. Explain the recovery of land revenue.
12. Explain the powers and functions of the Maharashtra Revenue Tribunal.

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#### *Unit II -- Maharashtra Tenancy and Agricultural Lands Act, 1948*

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 **Important Topics** • Historical Background, Object and Application of the Act • Deemed Tenant • Protected Tenant • Ceiling Area and Economic Holding • Irrigated Land • Maximum and Minimum Rent • Certificated Landlord • General Provisions Regarding Tenancies • Termination of Tenancies:

- Personal Cultivation
  - Non-Agricultural Use
  - By Certificated Landlord
  - Default
  - Surrender
- Relief Against Termination • Termination by Operation of Law • Tiller's Day • Purchase of Land by Tenants • Procedure for Purchase • Right of Tenant to Exchange Land • Management of Estates • Powers and Functions of Collector, Mamlatdar and Tribunal • Jurisdiction and Bar of Jurisdiction • Appeals and Revision

### ? Important Questions

1. Explain the historical background and objectives of the Act.
2. Define Deemed Tenant and Protected Tenant.
3. Explain Ceiling Area and Economic Holding.
4. Explain the rights and liabilities of tenants.
5. Explain the grounds for termination of tenancy.
6. What is Tiller's Day? Explain its significance.
7. Explain the procedure for purchase of land by tenants.
8. Explain the right of tenant to exchange land.
9. Explain the powers and functions of Collector and Mamlatdar.
10. Explain jurisdiction, appeals and revision under the Act.
11. Write short notes on: • Protected Tenant • Certificated Landlord • Economic Holding • Tiller's Day

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### Unit III -- Maharashtra Prevention of Fragmentation and Consolidation of Holdings Act, 1947

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✓ Important Topics • Historical Background, Object and Application • Determination of Local and Standard Areas • Entry in Record of Rights • Restrictions on Transfer and Partition of Fragments • Court Sales and Exceptions • Penalties • Transfer of Fragment to Government and Compensation • Consolidation of Holdings and Procedure • Reservation of Land for Public Purpose • Preparation and Enforcement of Scheme • Compensation and Apportionment • Certificate of Transfer • Rights in Holdings • Transfer of Encumbrances • Powers and Functions of:

- Consolidation Officer
  - Settlement Commissioner
- Bar of Jurisdiction

### ? Important Questions

1. Explain the objectives of the Fragmentation and Consolidation Act, 1947.
2. Explain local area and standard area.
3. Explain restrictions on transfer and partition of fragments.
4. Explain penalties under the Act.
5. Explain transfer of fragment to Government and compensation.
6. Explain the procedure of consolidation of holdings.
7. Explain reservation of land for public purposes.
8. Explain the preparation and enforcement of consolidation schemes.
9. Explain the powers and functions of the Consolidation Officer.
10. Explain the powers and functions of the Settlement Commissioner.
11. Explain the bar of jurisdiction under the Act.

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### Unit IV -- Ceiling Act, Mamlatdars' Courts Act and Land Acquisition Act

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✓ Important Topics  Maharashtra Agricultural Lands (Ceiling on Holdings) Act, 1961 • Historical Background and Object • Ceiling on Holdings of Land • Exempted Lands • Prohibition on Holding Land Beyond Ceiling Limits • Restrictions on Transfers and Acquisitions • Consequences of Contravention • Surplus Land • Submission of Returns • Selection and Determination of Surplus Land • Declaration of Surplus Land • Compensation • Distribution of Surplus Land

📌 *Mamlatdars' Courts Act, 1906* • Historical Background and Object • Mamlatdar and Joint Mamlatdar • Appointment and Powers • Cause of Action and Limitation • Injunction and Disobedience • Powers and Functions of Collector • Maharashtra Revenue Tribunal • Appeals

📌 *Land Acquisition Act* • Historical Background • Object and Application • Definitions under the Act

? *Important Questions*

1. Explain the objectives of the Ceiling on Holdings Act, 1961.
2. Explain exempted land and ceiling limits.
3. Explain restrictions on transfer and acquisition of land.
4. Explain the procedure for determination of surplus land.
5. Explain declaration and distribution of surplus land.
6. Explain the powers and functions of Mamlatdar.
7. Explain injunction and disobedience under the Mamlatdars' Courts Act.
8. Explain the powers of the Collector.
9. Explain the powers and functions of the Maharashtra Revenue Tribunal.
10. Explain the historical background and objectives of the Land Acquisition Act.
11. Write short notes on: • Surplus Land • Ceiling Area • Mamlatdar • Land Acquisition Act

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🔥 *Most Important for Exam (High Probability)*

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📖 *Theory* ✓ Powers and Duties of Revenue Officers ✓ Record of Rights ✓ Revenue Surveys and Boundary Marks ✓ Deemed and Protected Tenants ✓ Tiller's Day ✓ Purchase of Land by Tenants ✓ Consolidation of Holdings ✓ Ceiling on Holdings of Land ✓ Surplus Land and Compensation ✓ Powers of Mamlatdar and Maharashtra Revenue Tribunal

★ *Very Important Topics* ✓ Encroachment and Surrender of Land ✓ Rights in Unoccupied Lands ✓ Economic Holding and Ceiling Area ✓ Fragmentation and Consolidation Procedure ✓ Land Acquisition Act

🎯 *Most Repeated/Scoring Topics*

1. Record of Rights
2. Revenue Surveys
3. Deemed and Protected Tenant
4. Tiller's Day
5. Purchase of Land by Tenant
6. Consolidation of Holdings
7. Ceiling on Holdings and Surplus Land
8. Mamlatdar and MRT Powers
9. Land Acquisition Act

📖 *Project Management -- Important Topics & Questions (Exam-Oriented)*

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📖 *Unit I -- Introduction to Project Management*

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✅ *Important Topics* • Project Definition and Characteristics • Project Life Cycle • Processes and Knowledge Areas in Project Management • Work Breakdown Structure (WBS) and its Types • Introduction to PMBOK • Portfolio Management • Traditional vs Modern Project Management using PMBOK • Case Study: Online Shopping Project

? *Important Questions*

1. Define Project and explain its characteristics.
2. Explain the Project Life Cycle with a neat diagram.

3. Explain the processes and knowledge areas in Project Management.
4. What is Work Breakdown Structure (WBS)? Explain its types.
5. Explain the importance and benefits of WBS.
6. What is PMBOK? Explain its components.
7. Explain Portfolio Management.
8. Differentiate between Traditional and Modern Project Management.
9. Explain the PMBOK approach for project management.
10. Analyze the Online Shopping project using Project Management concepts.
11. Write short notes on: • WBS • PMBOK • Portfolio Management • Project Life Cycle

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## Unit II -- Agile Software Development

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 **Important Topics** • Introduction to Agile Development • Agile Methods • Scrum Framework • Agile vs Non-Agile Projects • Three Stages of Agile Project • Plan-Driven Development vs Agile Development • Extreme Programming (XP) • Scaling Agile Methods • Roles and Responsibilities in Agile • Scheduling and Tracking in Agile • Case Study: Online Shopping Project using Agile

### Important Questions

1. Explain the Agile Software Development methodology.
2. Explain various Agile methods.
3. Explain the Scrum framework.
4. Differentiate between Agile and Non-Agile projects.
5. Explain the three stages of an Agile project.
6. Differentiate between Plan-Driven and Agile Development.
7. Explain Extreme Programming (XP).
8. Explain the concept of Scaling Agile.
9. Explain various roles and responsibilities in Agile projects.
10. Explain scheduling and tracking in Agile.
11. Analyze the Online Shopping project using Agile methodology.
12. Create the three stages of the Online Shopping project using Agile.

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## Unit III -- Project Planning and Management

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 **Important Topics** • Introduction to Project Planning • Project Planning Process • Agile Project Management • Gantt Chart • PERT Chart • Critical Path Method (CPM) • Microsoft Project Software • Primavera Project Management Software • Role of Project Manager • Objectives of Activity Planning • Project Schedules • Activities, Sequencing and Scheduling • Project Planning Tools:

- Jira
- Kanban
- Extreme Programming

• Case Study: Develop a Software Project Plan

### Important Questions

1. Explain the Project Planning Process.
2. Explain Agile Project Management.
3. Explain Gantt Chart with an example.
4. Explain PERT Chart with an example.
5. Explain the Critical Path Method (CPM).
6. Differentiate between PERT and CPM.

7. Explain the role and responsibilities of a Project Manager.
8. Explain the objectives of Activity Planning.
9. Explain project scheduling and sequencing of activities.
10. Explain Microsoft Project and Primavera Software.
11. Develop a software project plan using any project management tool.
12. Write short notes on: • Gantt Chart • PERT • CPM • Jira and Kanban

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 *Unit IV -- Project Execution and Control*

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 **Important Topics • Project Execution:**

- Task Assignment
- Tracking
- Monitoring
- Project Control:
  - Schedule Control
  - Budget Control
  - Quality Control
- Earned Value Management (EVM) • Project Performance Measurement • Project Reporting and Communication • Risk Management Principles and Concepts • Risk Identification • Risk Analysis and Prioritization • Risk Response Planning and Implementation • Risk Monitoring and Review

 **Important Questions**

1. Explain Project Execution and its activities.
2. Explain task assignment, tracking and monitoring.
3. Explain schedule control, budget control and quality control.
4. What is Earned Value Management (EVM)? Explain its importance.
5. Explain project performance measurement.
6. Explain project reporting and communication.
7. Explain the principles and concepts of Risk Management.
8. Explain the process of risk identification.
9. Explain risk analysis and prioritization.
10. Explain risk response planning and implementation.
11. Explain risk monitoring and review.
12. Write short notes on: • EVM • Project Control • Risk Management • Project Reporting

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 **Most Important for Exam (High Probability)**

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 **Theory** ✓ Project Life Cycle ✓ Processes and Knowledge Areas in PM ✓ Work Breakdown Structure (WBS) ✓ PMBOK and Portfolio Management ✓ Agile Methodology and Scrum ✓ Agile vs Non-Agile Projects ✓ Gantt Chart, PERT and CPM ✓ Role of Project Manager ✓ Earned Value Management (EVM) ✓ Risk Management Process

 **Very Important Topics** ✓ Traditional vs Modern Project Management ✓ Three Stages of Agile Project ✓ Plan-Driven vs Agile Development ✓ Project Scheduling and Sequencing ✓ Project Reporting and Communication

 **Most Repeated/Scoring Topics**

1. Project Life Cycle
2. WBS and its Types
3. PMBOK and Knowledge Areas



4. Agile and Scrum
5. Agile vs Non-Agile Projects
6. Gantt Chart, PERT and CPM
7. Role of Project Manager
8. Earned Value Management (EVM)
9. Risk Identification and Risk Response Planning

 *Optimization Techniques -- Important Topics & Questions (Exam-Oriented)*

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 *Unit I -- Basic Concepts*

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 *Important Topics* • Historical Development of Optimization • Engineering Applications of Optimization • Formulation of Real-Life Problems as Mathematical Models • Steps in Optimization Problem Formulation • Objective Function and Constraints • Case Studies:

- Job Scheduling in IT Systems
- Cost Minimization in Cloud Services

 *Important Questions*

1. Define Optimization and explain its importance.
2. Explain the historical development of Optimization Techniques.
3. Discuss the engineering applications of optimization.
4. Explain the formulation of real-life problems as mathematical models.
5. Explain objective functions and constraints with examples.
6. Formulate a mathematical model for a given real-life problem.
7. Explain the optimization approach for job scheduling in IT systems.
8. Explain cost minimization in cloud services using optimization.
9. Write short notes on: • Mathematical Modelling • Objective Function • Constraints • Engineering Applications of Optimization

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 *Unit II -- Linear Programming*

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 *Important Topics* • Standard Form of LP Problem • Canonical Form of LP Problem • Assumptions of LP Models • Elementary Operations • Graphical Method for Two-Variable Problems • Simplex Method • Dual Simplex Method • Karmarkar's Projective Scaling Method • Case Studies:

- Resource Allocation in IT Projects
- Manpower Scheduling

 *Important Questions*

1. Define Linear Programming Problem (LPP).
2. Explain the standard and canonical forms of LPP.
3. Explain the assumptions of Linear Programming.
4. Solve an LPP using the graphical method. (*Numerical*)
5. Explain the Simplex Method.
6. Solve an LPP using the Simplex Method. (*Numerical*)
7. Explain the Dual Simplex Method.
8. Differentiate between Simplex and Dual Simplex Methods.
9. Explain Karmarkar's Projective Scaling Method.
10. Explain applications of Linear Programming in resource allocation and manpower scheduling.
11. Write short notes on: • LP Assumptions • Graphical Method • Simplex Method • Karmarkar's Method



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### Unit III -- Linear Programming Applications

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✓ **Important Topics** • Sensitivity Analysis • Transportation Problems • Assignment Problems • Solving Optimization Problems using Software • Graphical and Simplex Methods using Software Tools

? **Important Questions**

1. What is Sensitivity Analysis? Explain its importance.
2. Explain Transportation Problems.
3. Solve a Transportation Problem. (Numerical)
4. Explain Assignment Problems.
5. Solve an Assignment Problem. (Numerical)
6. Differentiate between Transportation and Assignment Problems.
7. Explain the use of software in solving Linear Programming Problems.
8. Explain graphical and simplex methods using software tools.
9. Write short notes on: • Sensitivity Analysis • Transportation Model • Assignment Model • Optimization Software

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### Unit IV -- Numerical Optimization Techniques

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✓ **Important Topics** • Basic Theory of Numerical Optimization • Method of Lagrange Multipliers • Karush-Kuhn-Tucker (KKT) Theory • Convex Optimization • Line Search Methods • Gradient Methods • Newton's Method • Conjugate Direction Methods • Quasi-Newton Methods • Projected Gradient Methods • Numerical Calculations using Software

? **Important Questions**

1. Explain the basic theory of Numerical Optimization.
2. Explain the Method of Lagrange Multipliers.
3. Solve problems using Lagrange Multipliers. (Numerical)
4. Explain the Karush-Kuhn-Tucker (KKT) Theory.
5. Explain Convex Optimization.
6. Explain Line Search Methods.
7. Explain Gradient Methods.
8. Explain Newton's Method.
9. Explain Conjugate Direction Methods.
10. Explain Quasi-Newton Methods.
11. Explain Projected Gradient Methods.
12. Explain the use of software for solving KKT problems.
13. Write short notes on: • Convex Optimization • KKT Theory • Newton's Method • Gradient Methods

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### Most Important for Exam (High Probability)

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 **Theory** ✓ Engineering Applications of Optimization ✓ Mathematical Model Formulation ✓ Standard and Canonical Forms of LPP ✓ Assumptions of Linear Programming ✓ Sensitivity Analysis ✓ Transportation and Assignment Problems ✓ Lagrange Multipliers ✓ Karush-Kuhn-Tucker (KKT) Theory ✓ Convex Optimization ✓ Newton's Method and Gradient Methods

 **Numericals (Very Important)** ✓ Graphical Method ✓ Simplex Method ✓ Dual Simplex Method ✓ Transportation Problem ✓ Assignment Problem ✓ Lagrange Multiplier Method ✓ KKT Numerical Problems

### *Most Repeated/Scoring Topics*

1. Graphical Method of LPP
2. Simplex Method
3. Transportation Problem
4. Assignment Problem
5. Sensitivity Analysis
6. Lagrange Multipliers
7. Karush-Kuhn-Tucker (KKT) Theory
8. Convex Optimization

### *Principles and Practices of Management -- Important Topics & Questions (Exam-Oriented)*

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#### *Unit I -- Introduction to Management and Organization*

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 *Important Topics* • Definition, Nature, Scope and Purpose of Management • Characteristics and Functions of Management • Evolution of Management Thought • Scientific Management • Modern Management • Principles of Management • Definition and Concept of Organization • Need and Purpose of Organization • Organizational Structure • Types of Organizations • Organizational Design

#### *Important Questions*

1. Define Management and explain its characteristics.
2. Explain the functions of Management.
3. Explain the evolution of Management thought.
4. Discuss Scientific Management and Modern Management.
5. Explain the principles of management.
6. Define Organization and explain its need and purpose.
7. Explain different types of organizational structures.
8. Differentiate between formal and informal organizations.
9. Explain various types of organizations.
10. Explain the importance of organizational structure.

#### *Open-Ended / Case Study Questions*

1. Study the organizational structure of your college or any company and explain the roles of each designation.
2. Design an organizational structure for a startup with 50 employees.
3. Analyze the management practices of an IT company or manufacturing firm.
4. Simulate the management structure required for a new e-commerce company.
5. Suggest management improvements for a struggling manufacturing company.

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#### *Unit II -- Manager, Leadership and Decision Making*

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 *Important Topics* • Manager and Managerial Roles • Skills of an Effective Manager • Functions of a Manager • Leadership: Meaning and Importance • Leadership Styles • Leadership Development • Leadership Behaviour • Nature and Process of Decision Making • Decision Making under Certainty and Uncertainty • Decision Making Process • Brainstorming

#### *Important Questions*

1. Who is a Manager? Explain the roles of a manager.
2. Explain the skills of an effective manager.
3. Explain the functions of a manager.

4. Define leadership and explain its importance.
5. Explain different leadership styles.
6. Explain leadership development and leadership behaviour.
7. Explain the decision-making process.
8. Differentiate between decision making under certainty and uncertainty.
9. Explain brainstorming as a decision-making technique.
10. Explain the relationship between leadership and decision making.

 *Open-Ended / Case Study Questions*

1. You are the team leader of a failing project. What leadership style would you adopt and why?
2. Simulate a decision-making process for launching a new product.
3. Analyze the leadership style of a famous CEO.
4. Suggest strategies to improve leadership skills among engineering students.
5. Prepare a case study on leadership and decision making in startups.

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 *Unit III -- Motivation and Communication*

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 *Important Topics* • Concept and Importance of Motivation • Classical Theories of Motivation • Modern Theories of Motivation • Financial and Non-Financial Motivation • Positive and Negative Motivation • Group Motivation • Definition and Meaning of Communication • Nature and Process of Communication • Types of Communication • Barriers to Communication

 *Important Questions*

1. Define Motivation and explain its importance.
2. Explain Classical and Modern theories of motivation.
3. Differentiate between financial and non-financial motivation.
4. Explain positive and negative motivation.
5. Explain group motivation.
6. Define communication and explain its nature.
7. Explain the communication process.
8. Explain various types of communication.
9. Explain barriers to communication.
10. Discuss methods to overcome communication barriers.

 *Open-Ended / Case Study Questions*

1. Design an employee motivation program for a startup.
2. Analyze the communication strategy used in a successful company.
3. Simulate a communication breakdown in a project and propose solutions.
4. Prepare a motivational plan for employees in a manufacturing industry.
5. Study the communication practices of an organization and suggest improvements.

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 *Unit IV -- Planning and Strategic Management*

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 *Important Topics* • Importance of Planning • Steps in Planning • Planning Premises • Types of Planning • Barriers to Effective Planning • Operational Planning • Strategic Planning • McKinsey's 7S Framework • SWOT Analysis • Meaning and Definition of Strategic Management • Elements of Strategic Management • Scope and Dimensions of Strategic Management • Strategic Management Process • Strategic Decisions

 *Important Questions*

1. Explain why management starts with planning.
2. Explain the steps involved in planning.
3. Explain planning premises.
4. Explain different types of planning.
5. Explain barriers to effective planning.
6. Differentiate between operational and strategic planning.
7. Explain McKinsey's 7S framework.
8. Explain SWOT analysis with examples.
9. Define Strategic Management and explain its importance.
10. Explain the strategic management process.
11. Explain strategic decisions and their importance.

#### *Open-Ended / Case Study Questions*

1. Prepare a SWOT analysis for your college or any startup.
2. Design a production planning system for a manufacturing company.
3. Simulate strategic planning for launching a new business.
4. Develop a strategic plan for an IT startup.
5. Analyze the strategic decisions taken by a successful company.
6. Prepare a demand forecasting and production scheduling plan for a manufacturing industry.



#### *Most Important for Exam (High Probability)*

 *Theory* ✓ Functions of Management ✓ Principles of Management ✓ Organizational Structure ✓ Leadership Styles ✓ Decision-Making Process ✓ Motivation Theories ✓ Communication Process and Barriers ✓ Planning Process ✓ SWOT Analysis ✓ Strategic Management Process ✓ McKinsey's 7S Framework

★ *Very Important Case Studies* ✓ Design an Organizational Structure for a Startup ✓ Decision-Making Process for a Business Problem ✓ Employee Motivation Program ✓ SWOT Analysis of a Company ✓ Production Planning System ✓ Strategic Plan for a Startup

#### *Most Repeated/Scoring Topics*

1. Functions and Principles of Management
2. Organizational Structure and Types of Organization
3. Leadership Styles and Managerial Roles
4. Decision-Making Process and Brainstorming
5. Motivation Theories and Communication Barriers
6. Planning Process and Types of Planning
7. SWOT Analysis and Strategic Management Process
8. McKinsey's 7S Framework
9. Operational Planning vs Strategic Planning
10. Strategic Decisions and Their Importance

#### *Financial Management -- IMP Topics & Questions (Exam Oriented)*



#### *Unit I -- Introduction to Financial Management*

✓ *IMP Topics* • Nature, Scope & Functions of Financial Management • Profit Maximization vs Wealth Maximization • Role of Finance Manager • Equity, Debt & Internal Financing • Importance of Finance for Engineers & Startups

#### ❓ *IMP Questions*

1. Define Financial Management and explain its functions.
2. Compare Profit Maximization and Wealth Maximization.
3. Explain the role of a Finance Manager.
4. Explain various sources of business finance.
5. Why is financial management important for engineers and startups?

 *Open-Ended / Case Study Questions* • Analyze the role of finance in a tech startup. • Explain financial decisions taken by engineering firms. • Suppose you are the CFO of a startup. How would you raise funds?

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## Unit II -- Time Value of Money & Capital Budgeting

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 *IMP Topics* • Time Value of Money (TVM) • Present Value (PV) • Future Value (FV) • Payback Period • Net Present Value (NPV) • Internal Rate of Return (IRR) • Accounting Rate of Return (ARR) • Profitability Index (PI) • Project Investment Evaluation

### *IMP Questions*

1. Explain the concept of Time Value of Money.
2. Explain Present Value and Future Value.
3. Explain Payback Period with example.
4. Explain NPV, IRR, ARR and PI methods.
5. Compare NPV and IRR methods.
6. Evaluate an engineering project using capital budgeting techniques.

 *Important Numericals* ✓ PV and FV ✓ Payback Period ✓ NPV ✓ IRR ✓ ARR ✓ PI

 *Open-Ended / Case Study Questions* • Calculate the NPV of a solar power plant project. • Determine the Payback Period for a manufacturing automation project. • A company has ₹50 lakh to invest in two projects. Which project should be selected and why?

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## Unit III -- Financial Statements & Ratio Analysis

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 *IMP Topics* • Profit & Loss Account • Balance Sheet • Cash Flow Statement • Liquidity Ratios • Profitability Ratios • Solvency Ratios • Turnover Ratios • Interpretation of Annual Reports

### *IMP Questions*

1. Explain Financial Statements and their importance.
2. Differentiate between P&L and Balance Sheet.
3. Explain Cash Flow Statement.
4. Explain Liquidity, Profitability and Solvency Ratios.
5. Explain Turnover Ratios.
6. How is Ratio Analysis useful in decision making?

 *Important Numericals* ✓ Current Ratio ✓ Quick Ratio ✓ Debt-Equity Ratio ✓ Inventory Turnover Ratio ✓ ROI

 *Open-Ended / Case Study Questions* • Perform ratio analysis of a listed engineering company. • Read and interpret the balance sheet of a manufacturing company. • Identify the financial strengths and weaknesses of a company using ratio analysis.

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## Unit IV -- Applications

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 *IMP Topics* • Working Capital and its Importance • Inventory, Receivables & Payables • Operating Cycle • Working Capital Requirement • Sources of Finance • Equity, Debt, Preference Shares &

Retained Earnings • Cost of Capital • Weighted Average Cost of Capital (WACC) • Role of Banks and Financial Institutions

 *IMP Questions*

1. Define Working Capital and explain its importance.
2. Explain the components of Working Capital.
3. Explain the Operating Cycle.
4. Explain various sources of finance.
5. Define Cost of Capital and WACC.
6. Explain the role of banks and capital markets.

 *Important Numericals* ✓ Working Capital Requirement ✓ Operating Cycle ✓ WACC Calculation

 *Open-Ended / Case Study Questions* • Analyze the working capital requirements of a construction project. • Design an inventory management strategy for a robotics company. • Suggest a funding mix for an electric vehicle startup. • Calculate WACC and recommend whether the project should be accepted. • As a finance manager, decide the best source of finance for a new startup.

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 *Most Important for Exam*

 *Theory* ✓ Profit vs Wealth Maximization ✓ Functions of Financial Management ✓ Role of Finance Manager ✓ Financial Statements & Ratio Analysis ✓ Working Capital ✓ Sources of Finance ✓ WACC

 *Numericals* ✓ PV & FV ✓ NPV ✓ IRR ✓ Payback Period ✓ Ratio Analysis ✓ Working Capital ✓ WACC

 *Most Important Open-Ended Questions*

1. Role of finance in a tech startup.
2. NPV of a solar plant project.
3. Ratio analysis of an engineering company.
4. Working capital analysis for a construction project.
5. Funding mix for an EV startup.
6. WACC-based investment decision.

 *Business Essentials for Rural Development -- IMP Topics & Questions (Exam Oriented)*

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 *Unit I -- Introduction to Business and Rural Economy*

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 *IMP Topics* • Meaning of Business • Types and Forms of Business Organizations • Role of Business in Economic Development • Indian Rural Economy and Demographics • Characteristics of Rural Markets and Consumers • Economic Activities in Rural Areas • Agriculture, Cottage Industries and Services • Challenges in Rural Business Development

 *IMP Questions*

1. Define Business and explain its types.
2. Explain the role of business in economic development.
3. Explain the characteristics of the Indian rural economy.
4. Explain the characteristics of rural consumers and markets.
5. Discuss the economic activities in rural areas.
6. Explain the challenges in rural business development.

 *Open-Ended / Case Study Questions* • Analyze the business opportunities available in your village. • Suggest solutions to improve rural infrastructure and business growth. • Prepare a report on

economic activities in a nearby rural area. • Identify a problem in a village and propose a business solution.

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 *Unit II -- Essentials of Entrepreneurship for Rural Development*

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✓ *IMP Topics* • Definition and Traits of an Entrepreneur • Importance of Rural Entrepreneurship • Steps in Setting Up a Small Business • Business Idea Generation • Opportunity Identification • Successful Rural Entrepreneurs and Enterprises • Social Entrepreneurship • Community-Driven Business Models

? *IMP Questions*

1. Define an entrepreneur and explain entrepreneurial traits.
2. Explain the importance of rural entrepreneurship.
3. Explain the steps involved in starting a rural business.
4. Explain business idea generation and opportunity identification.
5. Discuss social entrepreneurship and its importance.
6. Explain community-driven business models.

 *Open-Ended / Case Study Questions* • Design a small business plan for a village. • Study a successful rural entrepreneur and explain the reasons for success. • Suggest a startup idea to generate employment in rural areas. • Prepare a community-based entrepreneurship model for a village.

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 *Unit III -- Marketing, Finance and Operations in Rural Business*

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✓ *IMP Topics* • Marketing Mix (4Ps) • Rural Marketing Strategies • Communication, Pricing and Distribution • Basics of Financial Management • Cost, Revenue, Profit and Break-Even Analysis • Sources of Finance:

- Banks
- Microfinance
- SHGs
- NBFCs

• Operational Aspects:

- Procurement
- Inventory
- Quality
- Workforce

• Digital Tools and Mobile Technology in Rural Business

? *IMP Questions*

1. Explain the Marketing Mix (4Ps).
2. Explain rural marketing strategies.
3. Explain cost, revenue and profit.
4. What is Break-Even Analysis?
5. Explain various sources of finance for rural businesses.
6. Explain the role of SHGs and Microfinance.
7. Explain procurement and inventory management.
8. Explain the use of digital tools in rural business.



 *Open-Ended / Case Study Questions* • Prepare a marketing strategy for a rural product. • Design a financial plan for a village-based business. • Suggest how digital technologies can improve rural businesses. • Prepare a distribution strategy for selling products in rural areas.

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 *Unit IV -- Government Schemes, NGOs and Sustainable Development*

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 *IMP Topics* • Government Schemes:

- PMEGP
- MUDRA
- NRLM

• Role of NABARD • Role of KVIC • Role of NGOs in Rural Development • Sustainable Development Goals (SDGs) • Rural Empowerment • Engineering and Innovation for Rural Problems • Designing Rural Business Plans

 *IMP Questions*

1. Explain PMEGP, MUDRA and NRLM schemes.
2. Explain the role of NABARD in rural development.
3. Explain the role of KVIC.
4. Discuss the contribution of NGOs in rural development.
5. Explain Sustainable Development Goals (SDGs).
6. Explain the role of engineering and innovation in rural development.
7. Explain the importance of rural empowerment.

 *Open-Ended / Case Study Questions* • Design a sustainable business plan for a village. • Suggest engineering solutions for rural problems such as water, waste or energy. • Prepare a project report using a government scheme for starting a rural enterprise. • Analyze the impact of an NGO on rural development. • Design an innovative startup that contributes to Sustainable Development Goals (SDGs).

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 *Most Important for Exam*

 *Theory* ✓ Rural Economy and Rural Markets ✓ Rural Entrepreneurship ✓ Marketing Mix (4Ps) ✓ Sources of Finance ✓ Government Schemes (PMEGP, MUDRA, NRLM) ✓ NABARD and KVIC ✓ Sustainable Development Goals (SDGs) ✓ Rural Empowerment

 *Most Important Open-Ended Questions*

1. Design a rural business plan.
2. Study a successful rural entrepreneur.
3. Prepare a marketing strategy for a rural product.
4. Suggest digital solutions for rural businesses.
5. Design an engineering solution for a rural problem.
6. Prepare a sustainable development model for a village. 

 *Nanoscience and Nanotechnology -- IMP Topics & Questions (Exam Oriented)*

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 *Unit I -- Introduction to Nanoscience and Nanotechnology*

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 *IMP Topics* • Introduction to Nanoscale • Nanomaterials • Nanoscience and Nanotechnology • Nanoscale Bioassemblies • Liposomes • Viruses as Nanostructures • DNA Nanotechnology • Polysaccharides and Proteins • Protein Nanotubes • Nanofibers • Peptide Nanoparticles

 *IMP Questions*

1. Define Nanoscience and Nanotechnology.



2. Explain the concept of nanoscale.
3. What are nanomaterials? Explain their properties.
4. Explain nanoscale bioassemblies.
5. Explain the structure and applications of liposomes.
6. Explain DNA as a nanoscale material.
7. Explain protein nanotubes and nanofibers.
8. Explain peptide nanoparticles and their applications.

 *Open-Ended / Case Study Questions* • Why do materials exhibit different properties at the nanoscale? • Design a nanoscale system for targeted drug delivery. • Explain how biological molecules can be used in nanotechnology applications.

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 *Unit II -- Microbial Mediated Metallic Nanoparticles Synthesis*

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 *IMP Topics* • Microbial Synthesis of Nanoparticles • Gold Nanoparticles (AuNPs) • Silver Nanoparticles (AgNPs) • Au-Ag Alloy Nanoparticles • Oxide Nanoparticles • Magnetic Nanoparticles • Non-Magnetic Oxide Nanoparticles • Sulfide Nanoparticles

 *IMP Questions*

1. Explain microbial-mediated synthesis of nanoparticles.
2. Explain the synthesis and applications of Gold nanoparticles.
3. Explain the synthesis and applications of Silver nanoparticles.
4. Explain Au-Ag alloy nanoparticles.
5. Explain oxide nanoparticles.
6. Differentiate between magnetic and non-magnetic nanoparticles.
7. Explain sulfide nanoparticles and their applications.

 *Open-Ended / Case Study Questions* • Why are silver nanoparticles widely used as antibacterial agents? • Compare chemical and microbial synthesis of nanoparticles. • Design an eco-friendly method for synthesizing nanoparticles.

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 *Unit III -- Characterization Techniques for Nanomaterials*

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 *IMP Topics* • UV-Visible Spectroscopy • Fourier Transform Infrared Spectroscopy (FTIR) • X-Ray Diffraction (XRD) • X-Ray Photoelectron Spectroscopy (XPS) • Scanning Electron Microscopy (SEM) • Transmission Electron Microscopy (TEM) • Dynamic Light Scattering (DLS) • Vibrating Sample Magnetometer (VSM) • Introduction to SQUID

 *IMP Questions*

1. Explain UV-Visible Spectroscopy.
2. Explain the principle and applications of FTIR.
3. Explain XRD and its applications in nanomaterial characterization.
4. Explain XPS and its significance.
5. Explain the working of SEM.
6. Explain the working of TEM.
7. Explain Dynamic Light Scattering (DLS).
8. Explain Vibrating Sample Magnetometer (VSM).
9. Explain the principle of SQUID.

 *Open-Ended / Case Study Questions* • Which characterization technique would you choose to determine nanoparticle size and why? • Compare SEM and TEM for nanomaterial characterization. •

Design a characterization sequence for a newly synthesized nanoparticle.

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 *Unit IV -- Applications of Nanotechnology*

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✓ *IMP Topics* • Nanotechnology as an Antibacterial Agent • Drug Delivery Systems • Biosensors • Nanotechnology in Animal Industry • Nanotechnology in Wastewater Treatment • Biomedical Applications of Bioassemblies • Bioimaging • Vaccine Development

? *IMP Questions*

1. Explain the antibacterial applications of nanotechnology.
2. Explain nanotechnology-based drug delivery systems.
3. Explain biosensors and their applications.
4. Discuss the applications of nanotechnology in the animal industry.
5. Explain the role of nanotechnology in wastewater treatment.
6. Explain biomedical applications of bioassemblies.
7. Explain the applications of nanotechnology in bioimaging.
8. Explain the role of nanotechnology in vaccine development.

 *Open-Ended / Case Study Questions* • Design a nanoparticle-based drug delivery system for cancer treatment. • Explain how nanotechnology can solve water purification problems. • Suggest applications of nanotechnology in future healthcare systems. • Analyze the role of nanotechnology during vaccine development and disease control.

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 *Most Important for Exam*

 *Theory* ✓ Nanoscience and Nanotechnology Basics ✓ Nanoscale Bioassemblies ✓ Gold and Silver Nanoparticles ✓ Characterization Techniques (SEM, TEM, XRD, FTIR) ✓ Drug Delivery Systems ✓ Biosensors ✓ Wastewater Treatment ✓ Bioimaging and Vaccine Development

★ *Most Important Open-Ended Questions*

1. Design a nanoparticle-based drug delivery system.
2. Compare microbial and chemical synthesis of nanoparticles.
3. Select suitable characterization techniques for a nanoparticle.
4. Explain the role of nanotechnology in healthcare.
5. Propose a nanotechnology solution for water treatment.
6. Discuss future applications of nanotechnology in medicine and industry.  

 *Industrial Organization Management -- IMP Topics & Questions (Exam Oriented)*

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 *Unit I -- Management, Industrial Economics & Leadership*

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✓ *IMP Topics* • Meaning, Scope and Importance of Management • Functions of Management • Difference between Administration and Management • Leadership: Meaning and Importance • Types of Leadership:

- Autocratic
- Democratic
- Laissez-Faire

• Qualities of a Good Leader • Industrial Economics • Demand and Supply Concepts • Law of Demand and Law of Supply

? *IMP Questions*

1. Define Management and explain its functions.
2. Explain the scope and importance of management.

3. Differentiate between administration and management.
4. Define leadership and explain its importance.
5. Explain various types of leadership styles.
6. Explain the qualities of a good leader.
7. Define Industrial Economics.
8. Explain the concepts of demand and supply.
9. Explain the law of demand and the law of supply.

 *Open-Ended / Case Study Questions* • Which leadership style is best for a startup and why? • Analyze the leadership style of a successful CEO. • Explain how demand and supply affect the pricing of smartphones. • Simulate the management structure of a manufacturing company.

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 *Unit II -- Technology Management & Quality Management*

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 *IMP Topics* • Technology Management: Definition, Scope and Applications • Classification of Technology • Importance of Technology in National Economy • Quality Management and its Types • Quality of Design • Seven QC Tools • Poka Yoke (Mistake Proofing) • Quality Circles • Kaizen and its Implementation • Total Quality Management (TQM) • 5S System • Six Sigma • Toyota Case Study

 *IMP Questions*

1. Define Technology Management and explain its scope.
2. Explain the classification of technology.
3. Explain the importance of technology in the national economy.
4. Define Quality Management and explain its types.
5. Explain Quality of Design.
6. Explain the Seven QC Tools.
7. Explain Poka Yoke with examples.
8. Explain Quality Circles and their benefits.
9. Explain Kaizen and its implementation.
10. Explain TQM and Six Sigma.
11. Explain the 5S system.

 *Open-Ended / Case Study Questions* • Explain how Toyota uses the 5S system and Kaizen. • Design a quality improvement strategy for a manufacturing company. • Suggest methods to reduce defects in a production line using Poka Yoke. • Prepare a Six Sigma implementation plan for a factory.

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 *Unit III -- Marketing and Financial Management*

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 *IMP Topics* • Meaning of Market • Marketing Strategy and Motives • Types of Markets:

- Perfect Competition
- Monopoly
- Monopolistic Competition
- Oligopoly
- Online (Digital) Marketing • Financial Management • Types of Costs:
  - Fixed Cost
  - Variable Cost
  - Average Cost
  - Marginal Cost

- Total Cost
- Methods of Costing • Price and Capital • Debit and Credit • Profit and Loss Statement • Balance Sheet

### *IMP Questions*

1. Define Market and explain its characteristics.
2. Explain different market structures.
3. Differentiate between Monopoly and Perfect Competition.
4. Explain Online Marketing (Digital Marketing).
5. Define Financial Management.
6. Explain different types of costs.
7. Explain methods of costing.
8. Explain the concepts of debit and credit.
9. Explain Profit and Loss Statement.
10. Explain the Balance Sheet.

 *Open-Ended / Case Study Questions* • Analyze the market structure of the telecom industry in India. • Prepare a digital marketing strategy for a startup. • Prepare a Profit and Loss statement for a small business. • Analyze the cost structure of a manufacturing company.

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## *Unit IV -- Business Organization, Business Ownership & Entrepreneurship*

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 *IMP Topics* • Line Organization • Staff Organization • Functional Organization • Types of Business Ownership • One Person Company (OPC) • Sole Proprietorship • Partnership (Act 1934) • Limited Liability Partnership (LLP) Act, 2008 • Entrepreneurship and its Importance • Limitations of Rational Decision Making • Decision Making under:

- Certainty
- Uncertainty
- Risk
- Small and Medium Scale Industries (MSMEs) in India

### *IMP Questions*

1. Explain Line, Staff and Functional Organizations.
2. Explain different forms of business ownership.
3. Explain the concept of One Person Company (OPC).
4. Explain Sole Proprietorship and Partnership.
5. Explain Limited Liability Partnership (LLP).
6. Define Entrepreneurship and explain its importance.
7. Explain the limitations of rational decision making.
8. Differentiate between decision making under certainty, uncertainty and risk.
9. Explain the role of MSMEs in India.

 *Open-Ended / Case Study Questions* • Suggest the best form of business ownership for a technology startup. • Analyze the contribution of MSMEs to India's economy. • Design an organizational structure for a medium-scale industry. • Prepare a decision-making model for a new business under uncertain market conditions.

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## *Most Important for Exam*

 *Theory* ✓ Functions of Management ✓ Leadership Styles ✓ Demand and Supply Laws ✓ Technology Management ✓ Seven QC Tools, Kaizen, 5S, TQM, Six Sigma ✓ Market Structures ✓

Types of Costs ✓ Profit and Loss Statement & Balance Sheet ✓ Business Ownership Types ✓ Entrepreneurship and MSMEs

★ *Most Important Open-Ended Questions*

1. Leadership style for a startup.
2. Toyota's implementation of 5S and Kaizen.
3. Digital marketing strategy for a startup.
4. Cost analysis of a manufacturing company.
5. Best business ownership model for a new company.
6. Role of MSMEs in India's economic development. 🏠📚

📚 *Business Analytics -- IMP Topics & Questions (Exam Oriented)*

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📖 *Unit I -- Introduction to Business Analytics*

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✓ *IMP Topics* • Definition of Business Analytics • Types of Analytics:

- Descriptive Analytics
  - Diagnostic Analytics
  - Predictive Analytics
  - Prescriptive Analytics
- Concept of Insights • Business Analytics in Practice • Big Data and Importance of Data • Types of Data • Business Decision Modeling • Data vs Information vs Knowledge • Data Quality • 5Vs of Big Data • Big Data Collection and Ethics • Data Sources and Collection Methods • Data Privacy, Security and Ethics

? *IMP Questions*

1. Define Business Analytics and explain its importance.
2. Explain the types of Business Analytics.
3. Differentiate between Descriptive, Predictive and Prescriptive Analytics.
4. Explain the concept of insights in analytics.
5. What is Big Data? Explain its importance.
6. Explain the 5Vs of Big Data.
7. Differentiate between data, information and knowledge.
8. Explain data quality and its importance.
9. Explain data sources and collection methods.
10. Discuss privacy, security and ethical issues in Big Data.

🌐 *Open-Ended / Case Study Questions* • How can a company use Big Data to improve customer experience? • Analyze ethical issues involved in collecting customer data. • Suggest a data-driven solution for improving an online shopping platform.

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📖 *Unit II -- Analytical Decision-Making*

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✓ *IMP Topics* • Analytical Decision-Making Process • Characteristics of Analytical Decision-Making • Breaking Business Problems into Questions • Characteristics of Good Questions • Skills of a Good Business Analyst • Data Exploration and Visualization • Excel, Tableau and Power BI • Statistical Analysis and Hypothesis Testing (Conceptual) • Data Visualization • Exploratory Data Analysis (EDA) • Data Cleaning • Data Inspection

? *IMP Questions*

1. Explain the analytical decision-making process.
2. Explain the characteristics of analytical decision-making.

3. Explain the characteristics of good analytical questions.
4. Explain the skills required for a business analyst.
5. Explain data exploration and visualization.
6. Explain the role of Excel, Tableau and Power BI in analytics.
7. Explain the concept of statistical analysis and hypothesis testing.
8. Define Data Visualization and explain its importance.
9. Explain Exploratory Data Analysis (EDA).
10. Explain Data Cleaning and Data Inspection.

 *Open-Ended / Case Study Questions* • A company is facing declining sales. Frame analytical questions to solve the problem. • Design a dashboard for monitoring business performance. • Explain how Tableau or Power BI can help managers make decisions.

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### Unit III -- Predictive and Prescriptive Analytics

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 *IMP Topics* • Trend Lines • Regression Analysis:

- Linear Regression
- Multiple Regression
- Predictive Modeling • Forecasting Techniques • Data Mining and Business Intelligence • Data Mining Approaches • Data Exploration and Data Reduction • Classification • Association • Cause-Effect Modeling • Linear Optimization • Non-Linear Programming • Integer Optimization • Cutting Plane Algorithm • Decision Analysis • Risk and Uncertainty Methods • Text Analytics • Web Analytics

 *IMP Questions*

1. Explain Trend Analysis and Trend Lines.
2. Explain Linear and Multiple Regression.
3. Explain Predictive Modeling and Forecasting Techniques.
4. Define Data Mining and explain its approaches.
5. Explain the relationship between Data Mining and Business Intelligence.
6. Explain Classification and Association techniques.
7. Explain Cause-Effect Modeling.
8. Explain Linear and Non-Linear Optimization.
9. Explain Integer Optimization.
10. Explain Decision Analysis under Risk and Uncertainty.
11. Explain Text Analytics and Web Analytics.

 *Open-Ended / Case Study Questions* • Predict future sales of an e-commerce company using analytics techniques. • Suggest a data mining solution for fraud detection. • Explain how web analytics can improve an online business. • Design a predictive model for customer churn.

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### Unit IV -- Business Analytics in Marketing and Finance

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 *IMP Topics* • Marketing Analytics • Customer Segmentation • Targeting and Positioning (STP) • Campaign Management • ROI Measurement • Data-Driven Marketing Strategies • Financial Analytics • Risk Management • Credit Scoring • Financial Forecasting and Planning • Financial Performance Improvement through Analytics

 *IMP Questions*

1. Define Marketing Analytics and explain its importance.

2. Explain Customer Segmentation, Targeting and Positioning.
3. Explain Campaign Management and ROI Measurement.
4. Explain Data-Driven Marketing Strategies.
5. Explain Financial Analytics and its applications.
6. Explain Risk Management using analytics.
7. Explain Credit Scoring.
8. Explain Financial Forecasting and Planning.
9. Explain how analytics improves financial performance.

 **Open-Ended / Case Study Questions** • Design a customer segmentation strategy for an online shopping company. • Explain how analytics can improve a marketing campaign. • Design a credit scoring system for a bank. • Analyze how analytics can improve the financial performance of a company. • Suggest a data-driven marketing strategy for a startup.

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 **Most Important for Exam**

 **Theory** ✓ Types of Business Analytics ✓ Big Data and 5Vs ✓ Data Privacy and Ethics ✓ Analytical Decision-Making Process ✓ Data Visualization and EDA ✓ Regression Analysis ✓ Predictive Modeling and Forecasting ✓ Data Mining and Business Intelligence ✓ Marketing Analytics ✓ Financial Analytics

 **Most Important Open-Ended Questions**

1. Use Big Data to improve customer experience.
2. Frame analytical questions for a business problem.
3. Design a business dashboard using Tableau/Power BI.
4. Create a predictive model for sales or customer churn.
5. Design a customer segmentation strategy.
6. Explain how analytics improves financial performance. 

 **Mobile Journalism (MOJO) -- IMP Topics & Questions (Exam Oriented)**

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 **Unit I -- Mobile Journalism**

 **IMP Topics** • Definition of Mobile Journalism (MOJO) • Importance of Mobile Journalism • Need and Significance of MOJO • Mobile Generation • Influence of Mobile on Journalism • Future of Mobile Journalism • Web News Portals and E-Papers • OTT: Concept, Content Opportunities and Scope

 **IMP Questions**

1. Define Mobile Journalism and explain its importance.
2. Explain the need and significance of Mobile Journalism.
3. Discuss the influence of mobile technology on journalism.
4. Explain the future of Mobile Journalism.
5. Explain the concept of Web News Portals and E-Papers.
6. What is OTT? Explain its scope and opportunities.

 **Open-Ended / Case Study Questions** • Analyze how smartphones have changed journalism. • Compare traditional journalism and mobile journalism. • Explain the impact of OTT platforms on news consumption.

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 **Unit II -- MOJO and Applications**

 **IMP Topics** • Mobile Journalism Kits • Importance of Mobile Journalism Apps • Photo Editing Apps • Video Editing Apps • Audio Editing Apps • Global Adoption of MOJO • MOJO in India • Citizen Journalism • Blogs, Vlogs, Social Media and Podcasts



### ? IMP Questions

1. Explain the components of a Mobile Journalism Kit.
2. Explain the importance of Mobile Journalism Apps.
3. Explain the use of photo, video and audio editing apps.
4. Discuss the global adoption of MOJO.
5. Explain the growth of Mobile Journalism in India.
6. Define Citizen Journalism.
7. Explain the role of blogs, vlogs and podcasts in journalism.

 *Open-Ended / Case Study Questions* • Design a basic MOJO kit for a student journalist. • Explain how social media has empowered citizen journalism. • Analyze the role of podcasts and vlogs in modern journalism.

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### Unit III -- Basic Steps in Mobile Reporting

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 *IMP Topics* • Mobile Journalism Content • Mobile Apps for Reporting • Audio and Video Mobile Applications • Script Writing • Presentation Skills • Multimedia Content Management • Managing Social Media • Impact of Mobile on Society

### ? IMP Questions

1. Explain the process of Mobile Reporting.
2. Explain the use of mobile apps for reporting.
3. Explain audio and video applications used in MOJO.
4. Explain script writing for mobile journalism.
5. Explain presentation skills required in MOJO.
6. Explain multimedia content management.
7. Explain social media management.
8. Discuss the impact of mobile technology on society.

 *Open-Ended / Case Study Questions* • Prepare a script for reporting a local event using a smartphone. • Create a social media strategy for a news channel. • Analyze the impact of smartphones on society and journalism.

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### Unit IV -- Planning, Scripting, Shooting, Editing & Publishing

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 *IMP Topics* • Content Gathering in Mobile Journalism • News Gathering and Verification • Editing and Publishing • Technology and Apps for Content Creation • Media Platforms • Future of MOJO Reporting • Mobile Regulation Authorities:

- TRAI
- MIB
- AIMCR

• Ethics in Mobile Journalism • Citizen Journalism

### ? IMP Questions

1. Explain the process of content gathering in mobile journalism.
2. Explain news verification techniques.
3. Explain the role of technology in content creation and editing.
4. Explain different media platforms used in MOJO.
5. Discuss the future of Mobile Journalism.
6. Explain the functions of TRAI, MIB and AIMCR.



7. Explain ethics in Mobile Journalism.
8. Discuss the advantages and challenges of Citizen Journalism.

 *Open-Ended / Case Study Questions* • Cover a college event entirely using a smartphone and explain the process. • Explain how fake news can be prevented in mobile journalism. • Analyze ethical issues in citizen journalism. • Design a workflow for Planning -> Shooting -> Editing -> Publishing a news report.

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 *Most Important for Exam*

 *Theory* ✓ Definition and Importance of Mobile Journalism ✓ Future of MOJO ✓ OTT and E-Papers ✓ Mobile Journalism Apps and Kits ✓ Citizen Journalism ✓ Script Writing and Presentation Skills ✓ Multimedia Content Management ✓ Ethics in Mobile Journalism ✓ TRAI, MIB and AIMCR

★ *Most Important Open-Ended Questions*

1. Compare traditional journalism and Mobile Journalism.
2. Design a MOJO kit for a reporter.
3. Prepare a mobile reporting workflow for a live event.
4. Analyze the role of citizen journalism in society.
5. Explain methods to verify news and prevent fake news.
6. Create a mobile journalism project from planning to publishing. 

 *Property Law -- IMP Topics & Questions (Exam Oriented)*

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 *Unit I -- Concepts, Meaning and Types of Properties (Sections 1 to 9, Transfer of Property Act, 1882)* 

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✓ *IMP Topics* • Meaning and Definition of Property • Kinds of Property:

- Movable & Immovable
- Tangible & Intangible
- Existing & Future
- Real & Personal

• Doctrine of Notice • Meaning and Definition of Transfer of Property • Properties Which May Be Transferred • Essentials of Valid Transfer of Property • Operation and Methods of Transfer • Procedure for Transfer of Property

? *IMP Questions*

1. Define Property and explain its kinds.
2. Differentiate between movable and immovable property.
3. Explain tangible and intangible property.
4. Explain the Doctrine of Notice.
5. Define Transfer of Property.
6. Explain the properties that may be transferred.
7. Explain the essentials of a valid transfer of property.
8. Explain the methods of transfer of property.

 *Open-Ended / Case Study Questions* • Can future property be transferred? Justify with reasons. • A person purchases land without checking ownership documents. Discuss the applicability of the Doctrine of Notice. • Prepare the procedure for transferring a residential property.

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 *Unit II -- General Principles Relating to Transfer of Property (Sections 10 to 53A)* 

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✓ *IMP Topics* • Conditions Restraining Alienation • Transfer to Unborn Person • Rule Against Perpetuity • Direction for Accumulation • Vested Interest and Contingent Interest • Conditions

Precedent and Subsequent • Conditional Transfers • Doctrine of Election • Doctrine of Apportionment • Doctrine of Holding Out • Feeding the Grant by Estoppel • Doctrine of Lis Pendens • Fraudulent Transfer • Doctrine of Part Performance

 *IMP Questions*

1. Explain conditions restraining alienation.
2. Explain transfer to an unborn person.
3. Explain the Rule Against Perpetuity.
4. Differentiate between vested and contingent interest.
5. Explain conditions precedent and subsequent.
6. Explain the Doctrine of Election.
7. Explain the Doctrine of Apportionment.
8. Explain the Doctrine of Holding Out.
9. Explain Feeding the Grant by Estoppel.
10. Explain the Doctrine of Lis Pendens.
11. Explain fraudulent transfer.
12. Explain the Doctrine of Part Performance.

 *Open-Ended / Case Study Questions* • Can property be transferred to an unborn child? Explain with examples. • A seller transfers property while a court case is pending. Discuss the Doctrine of Lis Pendens. • Explain the consequences of a fraudulent transfer of property.

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 *Unit III -- Sale, Exchange and Mortgage (Sections 54 to 98 & Registration Act, 1908)*

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 *IMP Topics* • Meaning and Definition of Sale and Exchange • Distinction Between Sale and Exchange • Essentials of a Valid Sale • Parties to a Sale • Formalities of Sale • Sale vs Contract for Sale • Registration of Contract for Sale • Rights and Liabilities of Buyer and Seller • Exchange and Rights of Parties • Registration Act, 1908:

- Compulsory Registration
  - Optional Registration
  - Effect of Non-Registration
- Definition and Types of Mortgages • Mortgagor, Mortgagee and Mortgage Money • Essentials of Valid Mortgage • Rights and Liabilities of Mortgagor and Mortgagee • Charge, Mortgage, Pledge and Hypothecation • Doctrine of Substituted Security • Charge on Immovable Property • Marshalling

 *IMP Questions*

1. Define Sale and Exchange.
2. Differentiate between Sale and Exchange.
3. Explain the essentials of a valid sale.
4. Differentiate between Sale and Contract for Sale.
5. Explain the rights and liabilities of buyer and seller.
6. Explain compulsory and optional registration.
7. Explain the effects of non-registration.
8. Define Mortgage and explain its types.
9. Differentiate between mortgage, pledge and hypothecation.
10. Explain the rights and liabilities of mortgagor and mortgagee.
11. Explain Marshalling and Doctrine of Substituted Security.

 *Open-Ended / Case Study Questions* • Analyze the legal consequences of an unregistered sale deed. • Suggest the best type of mortgage for obtaining a housing loan. • Compare mortgage and hypothecation with practical examples.

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 *Unit IV -- Lease, Gift and Actionable Claims (Sections 105 to 137)*

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 *IMP Topics* • Definition of Lease • Lessor and Lessee • Premium and Rent • Kinds of Leases • Essentials of Valid Lease • Rights and Liabilities of Lessor and Lessee • Determination of Lease • Forfeiture and Relief Against Forfeiture • Leases for Agricultural Purposes • TPA vs Maharashtra Rent Control Act, 1999 • Definition of Gift • Donor and Donee • Essentials of Valid Gift • Revocation of Gifts • Donatio Mortis Causa • Gifts under Mohammedan Law • Actionable Claims:

- Definition
- Formalities
- Notice to Debtor
- Rights and Liabilities of Parties

 *IMP Questions*

1. Define Lease and explain its essentials.
2. Explain the rights and liabilities of lessor and lessee.
3. Explain determination and forfeiture of lease.
4. Differentiate between lease under TPA and Maharashtra Rent Control Act.
5. Define Gift and explain its essentials.
6. Explain revocation of gifts.
7. Differentiate between gift and donatio mortis causa.
8. Explain gifts under Mohammedan law.
9. Define Actionable Claims.
10. Explain formalities and importance of actionable claims.
11. Explain the requirement of notice to debtor.
12. Explain rights and liabilities of transferor and transferee.

 *Open-Ended / Case Study Questions* • Draft the essential clauses of a lease agreement. • Analyze whether a gift can be revoked after registration. • Explain the importance of actionable claims in modern commercial transactions. • Compare gifts under the Transfer of Property Act and Mohammedan Law.

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 *Most Important for Exam*

 *Theory* ✓ Kinds of Property ✓ Doctrine of Notice ✓ Rule Against Perpetuity ✓ Vested vs Contingent Interest ✓ Doctrine of Election & Lis Pendens ✓ Sale and Exchange ✓ Mortgage and its Types ✓ Registration Act, 1908 ✓ Lease and Determination of Lease ✓ Gift and Actionable Claims

 *Most Important Open-Ended Questions*

1. Applicability of Doctrine of Lis Pendens in property disputes.
2. Legal consequences of non-registration of documents.
3. Compare Mortgage, Pledge and Hypothecation.
4. Draft a lease agreement and explain its clauses.
5. Can a gift be revoked? Explain with examples.
6. Importance of actionable claims in commercial transactions.  